

रोल नं.

Roll No.

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परीक्षार्थी प्रश्न-पत्र कोड को उत्तर-पुस्तिका के मुख-पृष्ठ पर अवश्य लिखें।

Candidates must write the Q.P. Code on the title page of the answer-book.

विज्ञान SCIENCE

निर्धारित समय : 3 घण्टे

Time allowed : 3 hours

अधिकतम अंक : 80

Maximum Marks : 80

नोट	Note
(I) कृपया जाँच कर लें कि इस प्रश्न-पत्र में मुद्रित पृष्ठ 23 हैं।	(I) Please check that this question paper contains 23 printed pages.
(II) कृपया जाँच कर लें कि इस प्रश्न-पत्र में 39 प्रश्न हैं।	(II) Please check that this question paper contains 39 questions.
(III) प्रश्न-पत्र में दाहिने हाथ की ओर दिए गए प्रश्न-पत्र कोड को परीक्षार्थी उत्तर-पुस्तिका के मुख-पृष्ठ पर लिखें।	(III) Q.P. Code given on the right hand side of the question paper should be written on the title page of the answer-book by the candidate.
(IV) कृपया प्रश्न का उत्तर लिखना शुरू करने से पहले, उत्तर-पुस्तिका में प्रश्न का क्रमांक अवश्य लिखें।	(IV) Please write down the Serial Number of the question in the answer-book before attempting it.
(V) इस प्रश्न-पत्र को पढ़ने के लिए 15 मिनट का समय दिया गया है। प्रश्न-पत्र का वितरण पूर्वाह्न में 10.15 बजे किया जाएगा। 10.15 बजे से 10.30 बजे तक छात्र केवल प्रश्न-पत्र को पढ़ेंगे और इस अवधि के दौरान वे उत्तर-पुस्तिका पर कोई उत्तर नहीं लिखेंगे।	(V) 15 minute time has been allotted to read this question paper. The question paper will be distributed at 10.15 a.m. From 10.15 a.m. to 10.30 a.m., the students will read the question paper only and will not write any answer on the answer-book during this period.

सामान्य निर्देश :

निम्नलिखित निर्देशों को बहुत सावधानी से पढ़िए और उनका सख्ती से पालन कीजिए :

- (i) इस प्रश्न-पत्र में कुल 39 प्रश्न हैं। सभी प्रश्न अनिवार्य हैं।
- (ii) यह प्रश्न-पत्र पाँच खण्डों में विभाजित किया गया है - क, ख, ग, घ एवं ङ।
- (iii) खण्ड क - प्रश्न संख्या 1 से 20 तक बहुविकल्पीय प्रकार के प्रश्न हैं। प्रत्येक प्रश्न 1 अंक का है।
- (iv) खण्ड ख - प्रश्न संख्या 21 से 26 तक अति लघु-उत्तरीय प्रकार के प्रश्न हैं। प्रत्येक प्रश्न 2 अंकों का है। इन प्रश्नों के उत्तर 30 से 50 शब्दों में दिए जाने चाहिए।
- (v) खण्ड ग - प्रश्न संख्या 27 से 33 तक लघु-उत्तरीय प्रकार के प्रश्न हैं। प्रत्येक प्रश्न 3 अंकों का है। इन प्रश्नों के उत्तर 50 से 80 शब्दों में दिए जाने चाहिए।
- (vi) खण्ड घ - प्रश्न संख्या 34 से 36 तक दीर्घ-उत्तरीय प्रकार के प्रश्न हैं। प्रत्येक प्रश्न 5 अंकों का है। इन प्रश्नों के उत्तर 80 से 120 शब्दों में दिए जाने चाहिए।
- (vii) खण्ड ङ - प्रश्न संख्या 37 से 39 तक 3 स्रोत-आधारित/प्रकरण-आधारित इकाइयों के मूल्यांकन के चार-चार अंकों के प्रश्न (उप-प्रश्नों सहित) हैं।
- (viii) प्रश्न-पत्र में समग्र विकल्प नहीं दिया गया है। यद्यपि, कुछ खण्डों में आंतरिक विकल्प दिए गए हैं। इस प्रकार के प्रश्नों में केवल एक ही विकल्प का उत्तर दीजिए।

खण्ड - क

प्रश्न संख्या 1 से 20 में दिए गए चार विकल्पों में से सर्वाधिक उपयुक्त विकल्प का चयन कीजिए। गलत उत्तर के ऋणात्मक अंक नहीं है।

1. निम्नलिखित में से एक वियोजन (अपघटन) अभिक्रिया चुनिए जिसमें वियोजन के लिए ऊर्जा का स्रोत प्रकाश हो : 1
 - (a) $2\text{FeSO}_4 \rightarrow \text{Fe}_2\text{O}_3 + \text{SO}_2 + \text{SO}_3$
 - (b) $2\text{H}_2\text{O} \rightarrow 2\text{H}_2 + \text{O}_2$
 - (c) $2\text{AgBr} \rightarrow 2\text{Ag} + \text{Br}_2$
 - (d) $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$
2. जब किसी परखनली में दानेदार जिंक के कुछ टुकड़ों में 2 mL सोडियम हाइड्रॉक्साइड विलयन मिलाकर उसे गर्म किया जाता है, तो होने वाली रासायनिक अभिक्रिया को संतुलित रासायनिक समीकरण के रूप में इस प्रकार लिखा जा सकता है : 1
 - (a) $\text{NaOH} + \text{Zn} \rightarrow \text{NaZnO}_2 + \text{H}_2\text{O}$
 - (b) $2\text{NaOH} + \text{Zn} \rightarrow \text{Na}_2\text{ZnO}_2 + \text{H}_2$
 - (c) $2\text{NaOH} + \text{Zn} \rightarrow \text{NaZnO}_2 + \text{H}_2$
 - (d) $2\text{NaOH} + \text{Zn} \rightarrow \text{Na}_2\text{ZnO}_2 + \text{H}_2\text{O}$

General Instructions :

Read the following instructions very carefully and strictly follow them :

- (i) *This question paper comprises **39** questions. **All** questions are compulsory.*
- (ii) *This question paper is divided into **five** sections – **A, B, C, D** and **E**.*
- (iii) ***Section A** – Question Nos. **1** to **20** are multiple choice questions. Each question carries **1** mark.*
- (iv) ***Section B** – Question Nos. **21** to **26** are very short answer type questions. Each question carries **2** marks. Answer to these questions should be in the range of 30 to 50 words.*
- (v) ***Section C** – Question Nos. **27** to **33** are short answer type questions. Each question carries **3** marks. Answer to these questions should be in the range of 50 to 80 words.*
- (vi) ***Section D** – Question Nos. **34** to **36** are long answer type questions. Each question carries **5** marks. Answer to these questions should be in the range of 80 to 120 words.*
- (vii) ***Section E** – Question Nos. **37** to **39** are of 3 source-based/case-based units of assessment carrying **4** marks each with sub-parts.*
- (viii) *There is no overall choice. However, an internal choice has been provided in some sections. Only one of the alternatives has to be attempted in such questions.*

SECTION – A

Select and write the most appropriate option out of the four options given for each of the questions 1-20. There is no negative mark for the incorrect response.

1. Select from the following a decomposition reaction in which source of energy for decomposition is light : **1**
 - (a) $2\text{FeSO}_4 \rightarrow \text{Fe}_2\text{O}_3 + \text{SO}_2 + \text{SO}_3$
 - (b) $2\text{H}_2\text{O} \rightarrow 2\text{H}_2 + \text{O}_2$
 - (c) $2\text{AgBr} \rightarrow 2\text{Ag} + \text{Br}_2$
 - (d) $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$
2. When 2 mL of sodium hydroxide solution is added to few pieces of granulated zinc in a test tube and then warmed, the reaction that occurs can be written in the form of a balanced chemical equation as : **1**
 - (a) $\text{NaOH} + \text{Zn} \rightarrow \text{NaZnO}_2 + \text{H}_2\text{O}$
 - (b) $2\text{NaOH} + \text{Zn} \rightarrow \text{Na}_2\text{ZnO}_2 + \text{H}_2$
 - (c) $2\text{NaOH} + \text{Zn} \rightarrow \text{NaZnO}_2 + \text{H}_2$
 - (d) $2\text{NaOH} + \text{Zn} \rightarrow \text{Na}_2\text{ZnO}_2 + \text{H}_2\text{O}$

3. $\text{MnO}_2 + 4\text{HCl} \rightarrow \text{MnCl}_2 + 2\text{H}_2\text{O} + \text{Cl}_2$
 उपरोक्त अभिक्रिया रेडॉक्स अभिक्रिया है क्योंकि इसमें : 1
- (a) MnO_2 उपचयित और HCl अपचयित हो रहा है।
 (b) HCl उपचयित हो रहा है।
 (c) MnO_2 अपचयित हो रहा है।
 (d) MnO_2 अपचयित हो रहा है और HCl उपचयित हो रहा है।
4. नीचे दिए गए यौगिकों पर विचार कीजिए :
 FeSO_4 ; CuSO_4 ; CaSO_4 ; Na_2CO_3
 इनमें से किस यौगिक के क्रिस्टलीय रूप में एक अणु में क्रिस्टल जल के अणुओं की संख्या अधिकतम है ? 1
- (a) FeSO_4
 (b) CuSO_4
 (c) CaSO_4
 (d) Na_2CO_3
5. किसी तंत्रिका का वह स्थल जहाँ विद्युत आवेश रासायनिक सिगनल में परिवर्तित होता है उसे कहते हैं : 1
- (a) तंत्रिकाक्ष
 (b) द्रुमिका
 (c) तंत्रिकापेशीय संधि
 (d) कोशिका-काय
6. कक्ष ताप पर द्रव अवस्था में पायी जाने वाली एक धातु और एक अधातु क्रमशः हैं : 1
- (a) ब्रोमीन और मरकरी
 (b) मरकरी और आयोडीन
 (c) मरकरी और ब्रोमीन
 (d) आयोडीन और मरकरी
7. किसी बिम्ब को किसी उत्तल लेंस के सामने कितनी दूरी पर रखा जाना चाहिए ताकि पर्दे पर बिम्ब के समान साइज का प्रतिबिम्ब प्राप्त हो ? 1
- (a) लेंस की फोकस दूरी की दोगुनी दूरी से अधिक दूरी पर।
 (b) लेंस के मुख्य फोकस पर।
 (c) लेंस की फोकस दूरी की दोगुनी दूरी पर।
 (d) लेंस के प्रकाशिक केन्द्र और मुख्य फोकस के बीच।

3. $\text{MnO}_2 + 4\text{HCl} \rightarrow \text{MnCl}_2 + 2\text{H}_2\text{O} + \text{Cl}_2$
The reaction given above is a redox reaction because in this case : 1
- (a) MnO_2 is oxidised and HCl is reduced.
 - (b) HCl is oxidised.
 - (c) MnO_2 is reduced.
 - (d) MnO_2 is reduced and HCl is oxidised.
4. Consider the following compounds :
 FeSO_4 ; CuSO_4 ; CaSO_4 ; Na_2CO_3
The compound having maximum number of water of crystallisation in its crystalline form in one molecule is : 1
- (a) FeSO_4
 - (b) CuSO_4
 - (c) CaSO_4
 - (d) Na_2CO_3
5. In a nerve cell, the site where the electrical impulse is converted into a chemical signal is known as : 1
- (a) Axon
 - (b) Dendrites
 - (c) Neuromuscular junction
 - (d) Cell body
6. A metal and a non-metal that exists in liquid state at the room temperature are respectively : 1
- (a) Bromine and Mercury
 - (b) Mercury and Iodine
 - (c) Mercury and Bromine
 - (d) Iodine and Mercury
7. At what distance from a convex lens should an object be placed to get an image of the same size as that of the object on a screen ? 1
- (a) Beyond twice the focal length of the lens.
 - (b) At the principal focus of the lens.
 - (c) At twice the focal length of the lens.
 - (d) Between the optical centre of the lens and its principal focus.

8. कार्बन के यौगिक/यौगिकों :

- (i) विद्युत के सुचालक होते हैं।
- (ii) विद्युत के कुचालक होते हैं।
- (iii) के अणुओं के बीच प्रबल आकर्षण बल होता है।
- (iv) के अणुओं के बीच दुर्बल आकर्षण बल होता है।

इनमें सही कथन हैं :

1

- (a) (i) और (ii)
- (b) (ii) और (iii)
- (c) (ii) और (iv)
- (d) (i) और (iii)

9. ऐलुमिनियम और जिंक के ऑक्साइड होते हैं :

1

- (a) अम्लीय
- (b) क्षारकीय
- (c) उभयधर्मी
- (d) उदासीन

10. गुणसूत्र :

- (i) आनुवंशिक सूचना को जनकों से अगली संतति तक ले जाते हैं।
- (ii) किसी जन्तु कोशिका के केन्द्रक के भीतर धागेनुमा संरचना होती है।
- (iii) मानव जनन तंत्र में सदैव युग्मों में विद्यमान होते हैं।
- (iv) कोशिका विभाजन में सम्मिलित होते हैं।

इनमें सही कथन हैं :

1

- (a) (i) और (ii)
- (b) (iii) और (iv)
- (c) (i), (ii) और (iv)
- (d) (i) और (iv)

8. Carbon compounds :

- (i) are good conductors of electricity.
- (ii) are bad conductors of electricity.
- (iii) have strong forces of attraction between their molecules.
- (iv) have weak forces of attraction between their molecules.

The correct statements are :

1

- (a) (i) and (ii)
- (b) (ii) and (iii)
- (c) (ii) and (iv)
- (d) (i) and (iii)

9. Oxides of aluminium and zinc are :

1

- (a) acidic
- (b) basic
- (c) amphoteric
- (d) neutral

10. Chromosomes :

- (i) carry hereditary information from parents to the next generation.
- (ii) are thread like structures located inside the nucleus of an animal cell.
- (iii) always exist in pairs in human reproductive cells.
- (iv) are involved in the process of cell division.

The correct statements are :

1

- (a) (i) and (ii)
- (b) (iii) and (iv)
- (c) (i), (ii) and (iv)
- (d) (i) and (iv)

11. नीचे दिए गए कथनों पर विचार कीजिए :

- (i) किसी शिशु का लिंग इससे निर्धारित होता है कि उसे माता से क्या वंशानुगत हुआ है।
- (ii) किसी शिशु का लिंग इससे निर्धारित होता है कि उसे पिता से क्या वंशानुगत हुआ है।
- (iii) नर शिशु होने की प्रायिकता मादा शिशु होने की प्रायिकता से अधिक होती है।
- (iv) किसी शिशु का लिंग निर्धारण निषेचन के समय तब होता है जब नर युग्मक और मादा युग्मक के संलयन द्वारा युग्मनज बनता है।

इनमें सही कथन हैं :

1

- (a) (i) और (iii)
- (b) (ii) और (iv)
- (c) (iii) और (iv)
- (d) (i), (iii) और (iv)

12. निम्नलिखित में से कौनसा एक मादा जनन तंत्र का भाग नहीं है ?

1

- (a) अण्डाशय
- (b) गर्भाशय
- (c) शुक्रवाहिनी
- (d) फैलोपियन नलिका

13. नीचे दिए गए किस जीव की अलैंगिक जनन की विधि बहुखण्डन है ?

1

- (a) यीस्ट
- (b) लेस्मानिया
- (c) पैरामीशियम
- (d) प्लेज़मोडियम

14. ज़रा दूरदर्शिता के संशोधन के लिए उपयोग किए जाने वाले द्विफोकसी लेंसों का :

1

- (a) ऊपरी भाग उत्तल लेंस निकट दृष्टि के लिए और निचला भाग अवतल लेंस दूरदृष्टि के लिए होता है।
- (b) ऊपरी भाग उत्तल लेंस दूरदृष्टि के लिए और निचला भाग अवतल लेंस निकट दृष्टि के लिए होता है।
- (c) ऊपरी भाग अवतल लेंस निकट दृष्टि के लिए और निचला भाग उत्तल लेंस दूरदृष्टि के लिए होता है।
- (d) ऊपरी भाग अवतल लेंस दूरदृष्टि के लिए और निचला भाग उत्तल लेंस निकट दृष्टि के लिए होता है।

11. Consider the following statements :

- (i) The sex of a child is determined by what it inherits from the mother.
- (ii) The sex of a child is determined by what it inherits from the father.
- (iii) The probability of having a male child is more than that of a female child.
- (iv) The sex of a child is determined at the time of fertilisation when male and female gametes fuse to form a zygote.

The correct statements are :

1

- (a) (i) and (iii)
- (b) (ii) and (iv)
- (c) (iii) and (iv)
- (d) (i), (iii) and (iv)

12. Which one of the following organ is **NOT** a part of human female reproductive system ?

1

- (a) Ovary
- (b) Uterus
- (c) Vas deferens
- (d) Fallopian tube

13. In which of the following organisms, multiple fission is a means of asexual reproduction ?

1

- (a) Yeast
- (b) Leishmania
- (c) Paramoecium
- (d) Plasmodium

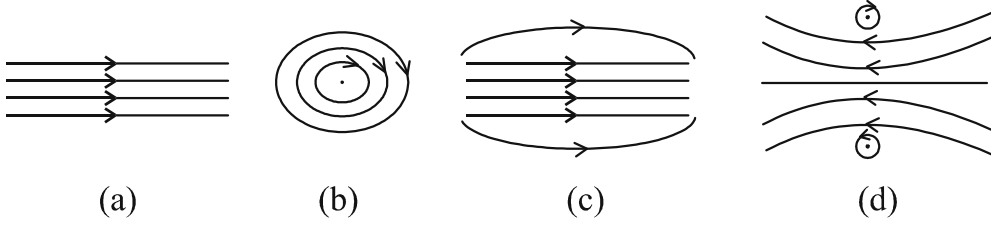
14. In bifocal lenses used for the correction of presbyopia :

1

- (a) the upper portion is of convex lens for the near vision and lower part is of concave lens for the distant vision.
- (b) the upper portion is of convex lens for the distant vision and lower part is of concave lens for the near vision.
- (c) the upper portion is of concave lens is for the near vision and lower part is of convex lens for the distant vision.
- (d) the upper portion is of concave lens for the distant vision and lower part is of convex lens for the near vision.

15. किसी धारावाही परिनालिका के भीतर उत्पन्न चुम्बकीय क्षेत्र का पैटर्न होता है :

1



16. निम्नलिखित में से वह आहार शृंखला पहचानिए जिसका द्वितीय पोषी स्तर का जीव विलुप्त है :

1

- (a) घास, बकरी, शेर
- (b) प्राणिप्लवक, पादपप्लवक, छोटी मछली, बड़ी मछली
- (c) बाघ, घास, सांप, मेंढक
- (d) टिड्डा, घास, सांप, मेंढक, गरुड़ (उकाब)

प्रश्न संख्या 17 से 20 में दो कथन- एक अभिकथन (A) और दूसरा कारण (R) दिया गया है। इन प्रश्नों के उत्तर नीचे दिए गए कोड (a), (b), (c) और (d) में से चुनकर दीजिए :

- (a) अभिकथन (A) और कारण (R) दोनों सही हैं और कारण (R), अभिकथन (A) की सही व्याख्या करता है।
- (b) अभिकथन (A) और कारण (R) दोनों सही हैं, किंतु कारण (R) अभिकथन (A) की सही व्याख्या नहीं करता है।
- (c) अभिकथन (A) सही है, परन्तु कारण (R) गलत है।
- (d) अभिकथन (A) गलत है, किंतु कारण (R) सही है।

17. अभिकथन (A) : इन्द्रधनुष आकाश में सूर्य के प्रकाश का प्राकृतिक स्पेक्ट्रम होता है।

कारण (R) : इन्द्रधनुष तभी बनता है जब आकाश में सूर्य सिर के ऊपर होता है तथा वायु में जल की सूक्ष्म बूंदें भी उपस्थित होती हैं।

1

18. अभिकथन (A) : जब जिंक नाइट्रिक अम्ल से अभिक्रिया करता है तो हाइड्रोजन गैस नहीं निकलती है।

कारण (R) : नाइट्रिक अम्ल अभिक्रिया में उत्पन्न हाइड्रोजन गैस को उपचयित करके पानी बनाता है और स्वयं अपचयित हो जाता है।

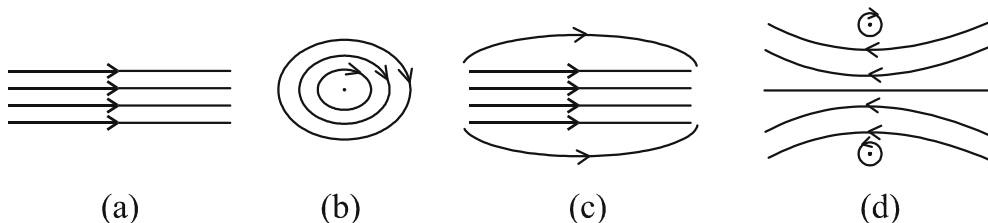
1

19. अभिकथन (A) : किसी आहार शृंखला के उच्चतम पोषी स्तर के जीवों में हानिकर रसायनों का संग्रहण अधिकतम होता है।

कारण (R) : फसलों को रोगों और पीड़कों से बचाने के लिए उन पर हानिकर रसायनों का छिड़काव किया जाता है।

1

15. The pattern of the magnetic field produced inside a current carrying solenoid is : 1



16. Identify the food chain in which the organisms of the second trophic level are missing : 1
- (a) Grass, goat, lion
 - (b) Zooplankton, Phytoplankton, small fish, large fish
 - (c) Tiger, grass, snake, frog
 - (d) Grasshopper, grass, snake, frog, eagle

For Q. Nos. 17 to 20, two statements are given – One labelled as **Assertion (A)** and the other labelled as **Reason (R)**. Select the correct answer to these questions from the codes (a), (b), (c) and (d) as given below :

- (a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).
- (b) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).
- (c) Assertion (A) is true, but Reason (R) is false.
- (d) Assertion (A) is false, but Reason (R) is true.

17. **Assertion (A) :** The rainbow is a natural spectrum of sunlight in the sky.
Reason (R) : Rainbow is formed in the sky when the sun is overhead and water droplets are also present in air. 1

18. **Assertion (A) :** Hydrogen gas is not evolved when zinc reacts with nitric acid.
Reason (R) : Nitric acid oxidises the hydrogen gas produced to water and itself gets reduced. 1

19. **Assertion (A) :** Accumulation of harmful chemicals is maximum in the organisms at the highest trophic level of a food chain.
Reason (R) : Harmful chemicals are sprayed on the crops to protect them from diseases and pests. 1

20. **अभिकथन (A) :** स्थलीय जीवों की तुलना में जलीय जीवों की श्वसन-दर अधिक तीव्र होती है।
- कारण (R) :** वायु में ऑक्सीजन की मात्रा की तुलना में पानी में घुली ऑक्सीजन की मात्रा बहुत अधिक होती है।
- 1

खण्ड – ख

21. (A) (i) मानव मस्तिष्क में परिधीय तंत्रिका तंत्र का महत्त्व लिखिए।
- (ii) मानव मस्तिष्क की आघातों और चोटों से सुरक्षा किस प्रकार होती है ?
- अथवा**
- (B) पादपों में रसायनों और जल की अनुक्रिया द्वारा वृद्धि के कारण दिशिक गतियों के नाम तथा प्रत्येक का एक-एक उदाहरण दीजिए।
22. (i) कारण दीजिए कि मांसाहारी जन्तुओं की अपेक्षा शाकाहारी जन्तुओं की क्षुद्रांत्र अधिक लम्बी क्यों होती है ?
- (ii) यद्यपि पेप्सिन और ट्रिप्सिन दोनों ही प्रोटीन पाचक एन्जाइम हैं फिर भी इनमें अन्तर होता है। एक उदाहरण देकर इस कथन की पुष्टि कीजिए।
23. नीचे दिए गए कथन को संतुलित रासायनिक समीकरण में परिवर्तित कीजिए :
- “जब बेरियम क्लोराइड ऐलुमिनियम सल्फेट के साथ अभिक्रिया करता है, तो ऐलुमिनियम क्लोराइड और बेरियम सल्फेट बनते हैं।”
- इस अभिक्रिया के प्रकार का उल्लेख अपने उत्तर की पुष्टि के लिए कारण देकर कीजिए।
24. (i) दो चुम्बकीय क्षेत्र रेखाएं एक दूसरे का प्रतिच्छेदन क्यों नहीं करती हैं ?
- (ii) किसी दिए गए प्रदेश में एकसमान चुम्बकीय क्षेत्र का निरूपण किस प्रकार किया जाता है ? अपने उत्तर की पुष्टि के लिए आरेख खींचिए।
25. किसी सीधे धारावाही चालक के कारण चुम्बकीय क्षेत्र रेखाओं के पैटर्न को खींचिए और इसमें चालक में धारा की दिशा और चुम्बकीय क्षेत्र रेखाओं की दिशा अंकित कीजिए।
26. कोई बिम्ब किसी 15 cm फोकस दूरी के उत्तल दर्पण से 10 cm दूरी पर स्थित है। दर्पण द्वारा बने बिम्ब के प्रतिबिम्ब की स्थिति ज्ञात कीजिए।

- 20. Assertion (A) :** The rate of breathing in aquatic organisms is much faster than in terrestrial organisms.
- Reason (R) :** The amount of oxygen dissolved in water is very high as compared to the amount of oxygen in air. **1**

SECTION – B

- 21. (A) (i)** Write the significance of peripheral nervous system in human beings. **2**
- (ii)** How is human brain protected from mechanical injuries and shocks ?

OR

- (B)** Name one directional growth movement each in response to chemicals and water in plants. Write an example for each of them. **2**
- 22. (i)** Give reason why herbivorous animals have longer, small intestine than carnivorous animals ? **2**
- (ii)** Although ‘Pepsin’ and ‘Trypsin’ are both protein digesting enzymes yet they differ from each other. Justify this statement by giving one difference between them.
- 23.** Translate the following statement into a balanced chemical equation.
“When barium chloride reacts with aluminium sulphate, aluminium chloride and barium sulphate are formed.”
State the type of this reaction giving reason to justify your answer. **2**
- 24. (i)** Two magnetic field lines do not intersect each other. Why ? **2**
- (ii)** How is a uniform magnetic field in a given region represented ?
Draw a diagram in support of your answer.
- 25.** Draw the pattern of the magnetic field lines due to a straight current carrying conductor indicating the direction of current in the conductor and the direction of the corresponding magnetic field lines. **2**
- 26.** An object is placed at a distance of 10 cm from a convex mirror of focal length 15 cm. Find the position of the image formed by the mirror. **2**

खण्ड – ग

27. (A) पादप → हरिण (मृग) → शेर

उपरोक्त आहार श्रृंखला में दूसरे पोषी स्तर के सभी जीवों को हटाने का प्रभाव पहले और तीसरे पोषी स्तर पर क्या होगा ? यदि तीसरे पोषी स्तर के सभी जीव आहार-जाल में उपस्थित होते तो उन पर क्या यही प्रभाव होता जो उपरोक्त आहार श्रृंखला में हुआ है ? पुष्टि कीजिए।

3

अथवा

(B) कोई गैस 'X' जो एक घातक विष है, वायुमण्डल के ऊपरी स्तर में पायी जाती है और एक आवश्यक प्रकार्य का संपादन करती है।

इस गैस का नाम और वायुमण्डल में इस गैस के प्रकार्य का उल्लेख कीजिए। इस गैस के स्तर के घटने से किस रसायन का नाम जुड़ा है ? इस गैस की परत की क्षति को रोकने के लिए किसी अन्तर्राष्ट्रीय संगठन द्वारा क्या कदम उठाए गए हैं ?

3

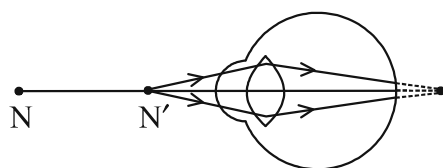
28. निम्नलिखित की दिशा को निर्धारित करने वाले नियम का नाम और नियम लिखिए :

3

- (i) किसी धारावाही सीधे चालक के चारों ओर उत्पन्न चुम्बकीय क्षेत्र की दिशा।
- (ii) किसी चुम्बकीय क्षेत्र में क्षेत्र के लम्बवत् स्थित धारावाही सीधे चालक पर आरोपित बल की दिशा।

29. नीचे दिए गए आरेख का अध्ययन करके निम्नलिखित प्रश्नों के उत्तर दीजिए:

3



- (i) इस आरेख द्वारा निरूपित दृष्टिदोष का नाम लिखिए। अपने उत्तर के लिए कारण दीजिए।
- (ii) इस दोष के दो कारणों की सूची बनाइए।
- (iii) आरेख की सहायता से यह दर्शाइए कि इस दृष्टिदोष का संशोधन किस प्रकार किया जाता है ?

30. प्रतिवर्ती क्रिया की परिभाषा लिखिए। प्रवाह आरेख की सहायता से छींकने जैसी प्रतिवर्ती क्रिया का पथ दर्शाइए।

3

SECTION – C

27. (A) Plants → Deer → Lion

In the given food chain, what will be the impact of removing all the organisms of second trophic level on the first and third trophic level ? Will the impact be the same for the organisms of the third trophic level in the above food chain if they were present in a food web ? Justify.

3

OR

(B) A gas 'X' which is a deadly poison is found at the higher levels of atmosphere and performs an essential function.

Name the gas and write the function performed by this gas in the atmosphere. Which chemical is linked to the decrease in the level of this gas ? What measures have been taken by an international organization to check the depletion of the layer containing this gas ?

3

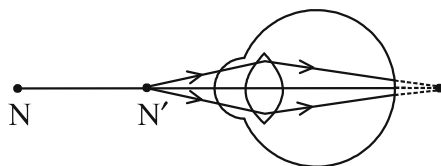
28. Name and state the rule to determine the direction of a :

3

- (i) magnetic field produced around a current carrying straight conductor.
- (ii) force experienced by a current carrying straight conductor placed in a magnetic field which is perpendicular to it.

29. Study the diagram given below and answer the questions that follow :

3



- (i) Name the defect of vision represented in the diagram. Give reason for your answer.
- (ii) List two causes of this defect.
- (iii) With the help of a diagram show how this defect of vision is corrected.

30. Define reflex action. With the help of a flow chart show the path of a reflex action such as sneezing.

3

31. (i) किन जीवों के हृदय में तीन कोष्ठ होते हैं ? इनके हृदय तीन कोष्ठों वाले क्यों होते हैं ? 3
 (ii) लसीका के दो कार्यों की सूची बनाइए।
32. किसी यौगिक, जिसे जिप्सम से बनाया जाता है, में यह गुण है कि वह उचित मात्रा में पानी मिलाए जाने पर कठोर हो जाता है : 3
 (i) इस यौगिक का सामान्य नाम और रासायनिक नाम लिखिए।
 (ii) इसके बनाने में होने वाली अभिक्रिया का रासायनिक समीकरण दीजिए।
 (iii) इसके दो उपयोगों की सूची बनाइए।
33. (i) वियोजन (अपघटन) अभिक्रिया की परिभाषा दीजिए। लैड नाइट्रेट को किसी क्वथन नली में अत्यधिक गर्म किए जाने पर होने वाली अभिक्रिया का रासायनिक समीकरण लिखिए। 3
 (ii) जल के विद्युत अपघटन में इलेक्ट्रॉडों पर दो गैसों मुक्त होती हैं। कैथोड और एनोड पर मुक्त होने वाली गैसों के द्रव्यमानों का अनुपात लिखिए।

खण्ड - घ

34. (A) (i) उल्लेख कीजिए कि क्या सभी बल्बों में धारा/विभवान्तर समान होंगे अथवा भिन्न-भिन्न होंगे जबकि तीन बल्बों को जिनकी : 5
 (a) वाटता समान है, श्रेणी में संयोजित किया गया है।
 (b) वाटता समान है, पार्श्व में संयोजित किया गया है।
 (c) वाटता भिन्न है, श्रेणी में संयोजित किया गया है।
 (d) वाटता भिन्न है, पार्श्व में संयोजित किया गया है।
 (ii) दो सर्वसम प्रतिरोधक जिनमें प्रत्येक का प्रतिरोध $24\ \Omega$ है, किसी 6 V की बैटरी से संयोजित है। इन प्रतिरोधकों के (a) निम्नतम प्रतिरोध के संयोजन और (b) अधिकतम प्रतिरोध के संयोजन में उपभुक्त शक्तियों का अनुपात परिकलित कीजिए।

अथवा

- (B) उस परिपथ का व्यवस्था आरेख खींचिए जिसमें 2 V के छः सेलों की बैटरी से एक $6\ \Omega$ का प्रतिरोधक, एक $12\ \Omega$ का प्रतिरोधक और एक $18\ \Omega$ का प्रतिरोधक तथा एक कुंजी सभी श्रेणी में संयोजित हैं। परिकलित कीजिए (जब कुंजी बन्द है) : 5
 (i) परिपथ में प्रवाहित विद्युत धारा
 (ii) $18\ \Omega$ के प्रतिरोधक के सिरों पर विभवान्तर
 (iii) $18\ \Omega$ के प्रतिरोधक में उपभुक्त शक्ति

31. (i) Which organisms have a three-chambered heart ? Why do they have three-chambered hearts ? 3
 (ii) List two functions of lymph.
32. A compound which is prepared from gypsum has the property of hardening when water is mixed in right quantity with it : 3
 (i) Write common name and the chemical name of this compound.
 (ii) Give chemical equation for its preparation.
 (iii) List its two uses.
33. (i) Define a decomposition reaction. Write chemical equation for the reaction that occurs when lead nitrate is heated strongly in a boiling tube. 3
 (ii) In electrolytic decomposition of water two gases are liberated at the electrodes. Give the mass ratio of the gas liberated at the cathode and at the anode.

SECTION – D

34. (A) (i) State whether the currents and potential difference in all the bulbs will be same or different when in a circuit three bulbs of : 5
 (a) same wattage are connected in series.
 (b) same wattage are connected in parallel.
 (c) different wattage are connected in series.
 (d) different wattage are connected in parallel.
 (ii) Two identical resistors of $24\ \Omega$ each are connected to a battery of 6 V . Calculate the ratio of the power consumed by the resulting combinations with (a) minimum resistance and (b) maximum resistance.

OR

- (B) Draw a schematic diagram of a circuit consisting of a battery of six 2 V cells, a $6\ \Omega$ resistor, a $12\ \Omega$ resistor and a $18\ \Omega$ resistor and a plug key all connected in series. Calculate the following (when key is closed) : 5
 (i) Electric current flowing in the circuit.
 (ii) Potential difference across $18\ \Omega$ resistor.
 (iii) Electric power consumed in $18\ \Omega$ resistor.

35. (A) (i) कार्बन यौगिकों की समजातीय श्रेणी की परिभाषा लिखिए। 5
(ii) C_4H_8 का गलनांक और क्वथनांक C_3H_6 अथवा C_2H_4 की तुलना में उच्चतर क्यों होता है ?
(iii) किसी समजातीय श्रेणी के यौगिकों के रासायनिक गुणधर्मों में क्रमबद्धता क्यों दृष्टिगोचर नहीं होती है ?
(iv) आण्विक सूत्र C_3H_6O के (i) ऐल्डिहाइड और (ii) कीटोन का नाम और संरचना लिखिए।

अथवा

- (B) (i) उस कार्बन के यौगिक 'X' का नाम और संरचना लिखिए जिसके अणु में दो कार्बन परमाणु हैं और जिसके साथ अनुलग्न $-OH$ जुड़ा है। 5
(ii) क्या होता है जब 'X' को आधिक्य सांद्र सल्फ्यूरिक अम्ल के साथ 443K पर गरम किया जाता है ? सभी आवश्यक शर्तों को दर्शाते हुए अभिक्रिया का रासायनिक समीकरण लिखिए तथा इस अभिक्रिया में सांद्र सल्फ्यूरिक अम्ल की भूमिका का उल्लेख भी कीजिए।
(iii) उपरोक्त अभिक्रिया में बनने वाले हाइड्रोकार्बन का नाम और उसकी इलेक्ट्रॉन-बिन्दु संरचना लिखिए।

36. (A) (i) मानव मादाओं के गर्भधारण को रोकने के लिए उपयोग की जाने वाली तीन तकनीकों/युक्तियों के नाम लिखिए। प्रत्येक के विपरीत प्रभाव का उल्लेख कीजिए। 5
(ii) क्या होगा यदि किसी मानव मादा में (a) निषेचन होता है, (b) अण्ड का निषेचन नहीं होता है ?

अथवा

- (B) (i) राइजोपस में बीजाणु समासंघ को आरेख खींचकर दर्शाइए तथा उन भागों को नामांकित कीजिए (a) जो जनन के भाग हैं और (b) जो जनन के भाग नहीं हैं। राइजोपस शुष्क रोटी के टुकड़े पर विकसित क्यों नहीं होते हैं ? 5
(ii) उस विधि का नाम और व्याख्या कीजिए जिसके द्वारा हाइड्रा जनन करता है।

- 35. (A)** (i) Define a homologous series of carbon compounds. **5**
(ii) Why is the melting and boiling points of C_4H_8 higher than that of C_3H_6 or C_2H_4 ?
(iii) Why do we **NOT** see any gradation in chemical properties of a homologous series compounds ?
(iv) Write the name and structures of (i) aldehyde and (ii) ketone with molecular form C_3H_6O .

OR

- (B)** (i) Write the name and structure of an organic compound 'X' having two carbon atoms in its molecule and its name is suffixed with '-ol'. **5**
(ii) What happens when 'X' is heated with excess concentrated sulphuric acid at 443 K ? Write chemical equation for the reaction stating the conditions for the reaction. Also state the role played by concentrated sulphuric acid in the reaction.
(iii) Name and draw the electron dot structure of hydrocarbon produced in the above reaction.

- 36. (A)** (i) Name three techniques/devices used by human females to avoid pregnancy. Mention the side effects caused by each. **5**
(ii) What will happen if in a human female (a) fertilisation takes place, (b) an egg is not fertilised ?

OR

- (B)** (i) Draw a diagram showing spore formation in Rhizopus and label the (a) reproductive and (b) non-reproductive parts. Why does Rhizopus not multiply on a dry slice of bread ? **5**
(ii) Name and explain the process by which reproduction takes place in Hydra.

खण्ड – ड

प्रश्न 37-39 स्रोत-आधारित/प्रकरण-आधारित प्रश्न हैं जिनके दो से तीन उप भाग हैं, जिनमें से एक उप भाग में आन्तरिक चयन प्रदान किया गया है :

37. मेंडल ने मटर के दिखाई देने वाले कई विपर्यासी लक्षणों का उपयोग करके आनुवंशिकता के नियमों पर कार्य किया। उन्होंने मटर के पौधों के एक अथवा दो विपर्यासी लक्षणों के युग्मों के साथ संकरण कराकर कई प्रयोग किए। अपने प्रेक्षणों के आधार पर उन्होंने कुछ व्याख्याएं कीं जिन्होंने आनुवंशिकता की क्रियाविधि के अध्ययन में सहायता की।

(i) जब मेंडल ने F_1 संतति प्राप्त करने के लिए मटर के शुद्ध लम्बे और शुद्ध बौने लक्षणों वाले पौधों के बीच संकरण कराया तो F_1 संतति के पौधों में उन्होंने कौनसे दो प्रेक्षण किए ?

1

(ii) प्रभावी और अप्रभावी लक्षणों के बीच एक अन्तर लिखिए।

1

(iii) (A) विपर्यासी लक्षणों के दो युग्मों के किसी संकरण में

$RRYY \times rryy$
(गोल पीले) (झुर्रीदार हरे)

मेंडल ने F_2 संतति में 4 प्रकार के संयोजनों का प्रेक्षण किया। किस विधि द्वारा उन्होंने F_2 संतति के पौधे प्राप्त किए थे ? प्राप्त जनकीय संयोजनों का अनुपात लिखिए तथा उल्लेख कीजिए कि इस प्रयोग से क्या निष्कर्ष निकलता है।

2

अथवा

(iii) (B) इस कथन की पुष्टि कीजिए :

“यह संभव है कि कोई लक्षण वंशानुगत तो हो जाए परन्तु व्यक्त न हो पाए।”

2

38. नीचे दिए गए आँकड़ों का अध्ययन कीजिए जिनमें तीन अवतल दर्पणों A, B और C की फोकस दूरी तथा उनके सामने स्थित बिम्ब की सापेक्ष दूरी दी गयी है :

प्रकरण	दर्पण	फोकस दूरी (cm)	बिम्ब-दूरी (cm)
1	A	20	45
2	B	15	30
3	C	30	20

(i) उपरोक्त प्रकरणों में से किसमें दर्पण द्वारा बने प्रतिबिम्ब का साइज बिम्ब के साइज से छोटा होगा ? अपने उत्तर की पुष्टि कीजिए।

1

(ii) प्रकरण 2 में बने प्रतिबिम्ब के दो गुणों की सूची बनाइए।

1

(iii) (A) दर्पण C द्वारा बने प्रतिबिम्ब की प्रकृति और साइज क्या है ? अपने उत्तर की पुष्टि के लिए किरण आरेख खींचिए।

2

SECTION – E

Q. No. 37-39 are source-based/case-based questions with 2 to 3 short sub-parts. Internal choice is provided in one of these sub-parts :

37. Mendel worked out the rules of heredity by working on garden pea using a number of visible contrasting characters. He conducted several experiments by making a cross with one or two pairs of contrasting characters of pea plant. On the basis of his observations he gave some interpretations which helped to study the mechanism of inheritance.

- (i) When Mendel crossed pea plants with pure tall and pure short characteristics to produce F_1 progeny, which two observations were made by him in F_1 plants ? 1
- (ii) Write one difference between dominant and recessive trait. 1
- (iii) (A) In a cross with two pairs of contrasting characters

$RRYY \quad \times \quad rryy$
(Round Yellow) (Wrinkled Green)

Mendel observed 4 types of combinations in F_2 generation. By which method did he obtain F_2 generation ? Write the ratio of the parental combinations obtained and what conclusions were drawn from this experiment. 2

OR

- (iii) (B) Justify the statement :
“It is possible that a trait is inherited but may not be expressed.” 2

38. Study the data given below showing the focal length of three concave mirrors A, B and C and the respective distances of objects placed in front of the mirrors :

Case	Mirror	Focal Length (cm)	Object Distance (cm)
1	A	20	45
2	B	15	30
3	C	30	20

- (i) In which one of the above cases the mirror will form a diminished image of the object ? Justify your answer. 1
- (ii) List two properties of the image formed in case 2. 1
- (iii) (A) What is the nature and size of the image formed by mirror C ?
Draw ray diagram to justify your answer. 2

अथवा

- (iii) (B) कोई बिम्ब 12 cm फोकस दूरी के अवतल दर्पण के ध्रुव से 18 cm की दूरी पर स्थित है। इस प्रकरण में प्रतिबिम्ब की स्थिति ज्ञात कीजिए। 2

39. अपचयन की विभिन्न प्रक्रियाओं द्वारा उत्पन्न धातुएं बहुत शुद्ध नहीं होती हैं। इनमें अशुद्धियाँ होती हैं। शुद्ध धातुएं प्राप्त करने के लिए इन अशुद्धियों को दूर किया जाता है। धातुओं को शुद्ध करने की सबसे अधिक प्रचलित विधि विद्युत अपघटनी परिष्करण है।

- (i) इस प्रक्रिया द्वारा कॉपर के परिष्करण के लिए कैथोड और ऐनोड किसके बने होते हैं ? 1
- (ii) उपरोक्त प्रक्रिया में उपयोग किए जाने वाले विलयन का नाम और उसका सूत्र लिखिए। 1
- (iii) (A) जब इस विद्युत अपघटनी सेल में विद्युत धारा प्रवाहित की जाती है तो कॉपर का परिष्करण किस प्रकार हो जाता है ? 2

अथवा

- (iii) (B) आपके पास दो बीकरों 'A' और 'B' में कॉपर सल्फेट विलयन भरा है। यदि आप बीकर 'A' में जिंक की पत्री और बीकर 'B' में सिल्वर की पत्री डुबो दें तो लगभग दो घन्टे के पश्चात् आप क्या प्रेक्षण करेंगे। प्रत्येक प्रकरण के प्रेक्षण का कारण दीजिए। 2

OR

- (iii) (B) An object is placed at a distance of 18 cm from the pole of a concave mirror of focal length 12 cm. Find the position of the image formed in this case. **2**

39. The metals produced by various reduction processes are not very pure. They contain impurities, which must be removed to obtain pure metals. The most widely used method for refining impure metals is electrolytic refining.

- (i) What is the cathode and anode made of in the refining of copper by this process ? **1**
- (ii) Name the solution used in the above process and write its formula. **1**
- (iii) (A) How copper gets refined when electric current is passed in the electrolytic cell ? **2**

OR

- (iii) (B) You have two beakers 'A' and 'B' containing copper sulphate solution. What would you observe after about 2 hours if you dip a strip of zinc in beaker 'A' and a strip of silver in beaker 'B' ? Give reason for your observations in each case. **2**

Marking Scheme
Strictly Confidential
Secondary School Examination, 2024
SUBJECT NAME SCIENCE (086) (Q.P. CODE 31/1/1)

General Instructions: -

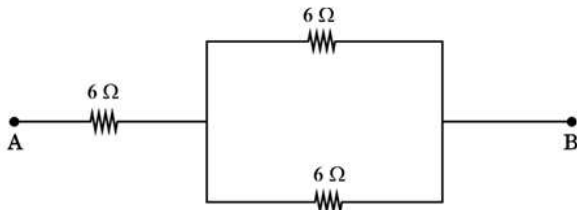
1	You are aware that evaluation is the most important process in the actual and correct assessment of the candidates. A small mistake in evaluation may lead to serious problems which may affect the future of the candidates, education system and teaching profession. To avoid mistakes, it is requested that before starting evaluation, you must read and understand the spot evaluation guidelines carefully.
2	“Evaluation policy is a confidential policy as it is related to the confidentiality of the examinations conducted, Evaluation done and several other aspects. Its’ leakage to public in any manner could lead to derailment of the examination system and affect the life and future of millions of candidates. Sharing this policy/document to anyone, publishing in any magazine and printing in News Paper/Website etc may invite action under various rules of the Board and IPC.”
3	Evaluation is to be done as per instructions provided in the Marking Scheme. It should not be done according to one’s own interpretation or any other consideration. Marking Scheme should be strictly adhered to and religiously followed. However, while evaluating, answers which are based on latest information or knowledge and/or are innovative, they may be assessed for their correctness otherwise and due marks be awarded to them. In class-X, while evaluating two competency-based questions, please try to understand given answer and even if reply is not from marking scheme but correct competency is enumerated by the candidate, due marks should be awarded.
4	The Marking scheme carries only suggested value points for the answers These are in the nature of Guidelines only and do not constitute the complete answer. The students can have their own expression and if the expression is correct, the due marks should be awarded accordingly.
5	The Head-Examiner must go through the first five answer books evaluated by each evaluator on the first day, to ensure that evaluation has been carried out as per the instructions given in the Marking Scheme. If there is any variation, the same should be zero after deliberation and discussion. The remaining answer books meant for evaluation shall be given only after ensuring that there is no significant variation in the marking of individual evaluators.
6	Evaluators will mark(✓) wherever answer is correct. For wrong answer CROSS ‘X’ be marked. Evaluators will not put right (✓)while evaluating which gives an impression that answer is correct and no marks are awarded. This is most common mistake which evaluators are committing.
7	If a question has parts, please award marks on the right-hand side for each part. Marks awarded for different parts of the question should then be totaled up and written in the left-hand margin and encircled. This may be followed strictly.
8	If a question does not have any parts, marks must be awarded in the left-hand margin and encircled. This may also be followed strictly.
9	If a student has attempted an extra question, answer of the question deserving more marks should be retained and the other answer scored out with a note “Extra Question”.
10	No marks to be deducted for the cumulative effect of an error. It should be penalized only once.
11	A full scale of marks 0 – 80 (example 0 to 80/70/60/50/40/30 marks as given in Question Paper) has to be used. Please do not hesitate to award full marks if the answer deserves it.

12	Every examiner has to necessarily do evaluation work for full working hours i.e., 8 hours every day and evaluate 20 answer books per day in main subjects and 25 answer books per day in other subjects (Details are given in Spot Guidelines). This is in view of the reduced syllabus and number of questions in question paper.
13	Ensure that you do not make the following common types of errors committed by the Examiner in the past:- Leaving answer or part thereof unassessed in an answer book. Giving more marks for an answer than assigned to it. Wrong totaling of marks awarded on an answer. Wrong transfer of marks from the inside pages of the answer book to the title page. Wrong question wise totaling on the title page. Wrong totaling of marks of the two columns on the title page. Wrong grand total. Marks in words and figures not tallying/not same. Wrong transfer of marks from the answer book to online award list. Answers marked as correct, but marks not awarded. (Ensure that the right tick mark is correctly and clearly indicated. It should merely be a line. Same is with the X for incorrect answer.) Half or a part of answer marked correct and the rest as wrong, but no marks awarded.
14	While evaluating the answer books if the answer is found to be totally incorrect, it should be marked as cross (X) and awarded zero (0) Marks.
15	Any unassessed portion, non-carrying over of marks to the title page, or totaling error detected by the candidate shall damage the prestige of all the personnel engaged in the evaluation work as also of the Board. Hence, in order to uphold the prestige of all concerned, it is again reiterated that the instructions be followed meticulously and judiciously.
16	The Examiners should acquaint themselves with the guidelines given in the “Guidelines for Spot Evaluation” before starting the actual evaluation.
17	Every Examiner shall also ensure that all the answers are evaluated, marks carried over to the title page, correctly totaled and written in figures and words.
18	The candidates are entitled to obtain photocopy of the Answer Book on request on payment of the prescribed processing fee. All Examiners/Additional Head Examiners/Head Examiners are once again reminded that they must ensure that evaluation is carried out strictly as per value points for each answer as given in the Marking Scheme.

MARKING SCHEME
Secondary School Examination, 2024
SCIENCE (Subject Code–086)
[Paper Code: 31/1/1]

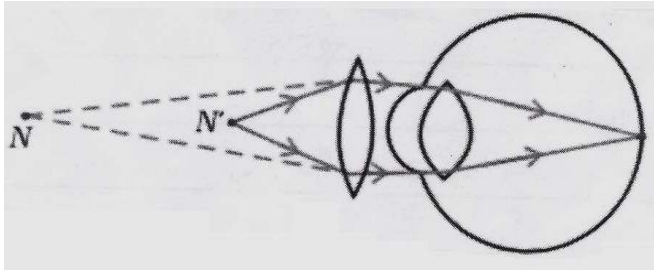
Maximum Marks: 80

Q. No.	EXPECTED ANSWER / VALUE POINTS	Marks	Total Marks
SECTION A			
1	(b) / $2 \text{NaOH} + \text{Zn} \rightarrow \text{Na}_2\text{ZnO}_2 + \text{H}_2$	1	1
2	(c) / $2 \text{AgBr} \rightarrow 2 \text{Ag} + \text{Br}_2$	1	1
3	(c) /Mercury and Bromine	1	1
4	(c) / (ii) and (iv)	1	1
5	(d) / Na_2CO_3	1	1
6	(c) /amphoteric	1	1
7	(d) / MnO_2 is reduced and HCl is oxidised	1	1
8	(b) / (ii) and (iv)	1	1
9	(d) / (i) and (iv)	1	1
10	(c) /Neuromuscular junction	1	1
11	(c) / (ii) and (iii)	1	1
12	(c) /At twice the focal length of the lens	1	1
13	(d) /Retina	1	1
14	(a) / 	1	1
15	(c) /Tiger, grass, snake, frog	1	1
16	(d) / Plasmodium	1	1
17	(a) /Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).	1	1
18	(b) / Both Assertion (A) and Reason (R) are true, but Reason (R) is <i>not</i> the correct explanation of Assertion (A).	1	1
19	(c) /Assertion (A) is true, but Reason (R) is false.	1	1
20	(c) /Assertion (A) is true, but Reason (R) is false.	1	1
SECTION B			
21	Combination reaction – Single product is formed (or any other) $\text{CaO(s)} + \text{H}_2\text{O(l)} \rightarrow \text{Ca(OH)}_2\text{(aq)} + \text{Heat}$ Quick lime Slaked lime/Calcium hydroxide	$\frac{1}{2} + \frac{1}{2}$ 1	2
22	Role of: (i) Hydrochloric acid: Creates an acidic medium for facilitating the action of enzyme / kills microorganisms. (ii) Villi: Increases the surface area for absorption of digested food. (iii) Anal Sphincter: Exit of waste material from anus is regulated.	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$	

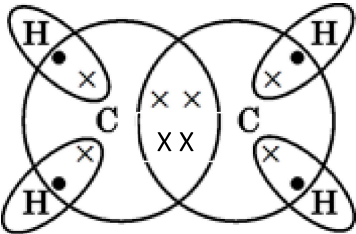
	(iv) Lipase: Breakdown / digestion of emulsified fats or lipids	½	2	
23	Movement of leaves of sensitive plant (i) Stimulus is touch. (ii) No growth is involved in the movement. (iii) Non directional Downward movement of roots Stimulus is gravity. Growth is involved in the movement Directional (Any two) (Any other suitable difference) OR (B) • Thyroxine • Thyroid gland • Iodine is necessary for thyroid gland to make thyroxine hormone. Deficiency of iodine in our diet causes goitre. <td>1+1</td> <td> ½ ½ 1 </td> <td>2</td>	1+1	½ ½ 1	2
24	u = - 10cm; f = +15 cm $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$ $\frac{1}{15} = \frac{1}{v} + \frac{1}{-10\text{ cm}}$ $\frac{1}{v} = \frac{1}{15\text{ cm}} + \frac{1}{10\text{ cm}}$ $v = + 6\text{ cm}$ Image is formed behind the mirror.	½ ½ 1	2	
25	(A) When two 6 Ω resistances are connected in parallel and the third resistance of 6Ω is connected in series combinations to this, then equivalent resistance will be 9 Ω / 	1		

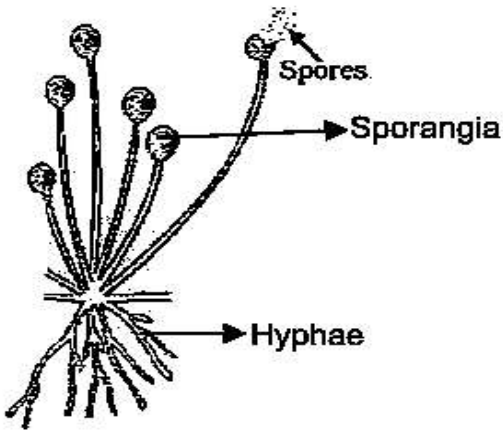
	[Award marks for writing the statement or drawing the diagram] $\frac{1}{R_p} = \frac{1}{6\ \Omega} + \frac{1}{6\ \Omega}$ $\therefore R_p = 3\ \Omega$ $R_s = 6 + 3 = 9\ \Omega$ OR (B) Equivalent resistance = $R_1 + R_2 = 1\ \Omega + 2\ \Omega = 3\ \Omega$ $I = \frac{V}{R}$ $= \frac{6\ V}{1\ \Omega + 2\ \Omega} = \frac{6\ V}{3\ \Omega} = 2\ A$ Electric power, $P = I^2 R$ $= (2A)^2 \times 2\ \Omega = 4 \times 2\ W = 8\ W$	1	
		1/2	
		1/2	
		1/2	
		1/2	
			2
26	(i) If they intersect then at the point of intersection, there would be two directions of magnetic field or compass needle would point towards two directions, which is not possible. (ii) Uniform magnetic field is represented by equidistant parallel straight lines 	1	
		1/2	
		1/2	
			2
	SECTION C		
27	(i) Change in colour: The solution will become green in colour. $\text{Fe(s)} + \text{CuSO}_4(\text{aq}) \longrightarrow \text{FeSO}_4 + \text{Cu(s)}$ Blue Green (or any other reaction which shows change in colour) (ii) Change in temperature: The temperature will increase. $\text{NaOH(aq)} + \text{HCl(aq)} \longrightarrow \text{NaCl(aq)} + \text{H}_2\text{O(l)} + \text{Heat}$	1/2	
		1/2	
		1/2	
		1/2	

	(or any other reaction which shows change in temperature) (iii) Formation of precipitate: Yellow precipitate of PbI_2 is formed. $\text{Pb}(\text{NO}_3)_2 (\text{aq}) + 2 \text{KI}(\text{aq}) \longrightarrow \text{PbI}_2(\text{s}) + 2 \text{KNO}_3(\text{aq})$ Yellow (or any other reaction which shows formation of precipitate)	$\frac{1}{2}$ $\frac{1}{2}$	3
28	(i) The taste of tomato juice will be slightly sour; The pH 4.6 indicates that tomato juice is an acid and acids are sour in taste. (ii) Acids that give more H^+ ions / H_3O^+ are Strong Acids Bases that give less OH^- ions are Weak Bases. (iii) Living animals can survive within a pH range of 7.0 to 7.8. So, if the pH of river water becomes low due to acid rain (pH < 5.6), then survival of aquatic animals becomes difficult.	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ 1	3
29	(i) Diffusion /Diffusion pressure alone cannot take care of oxygen delivery to all parts of the body. (ii) Reasons: (a) To ensure that the air-passage does not collapse. (b) There is sufficient time for oxygen to be absorbed and for the carbon dioxide to be released. (c) Chest cavity becomes larger. (d) Because exchange of gases takes place in the alveoli.	1 $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$	3
30	Reflex action is a sudden/spontaneous/immediate action in response to the environment/stimulus e.g. sneezing. Stimulus $\longrightarrow$ Receptors (Nose) $\longrightarrow$ Sensory neuron $\longrightarrow$ Spinal cord $\longrightarrow$ Motor neuron $\longrightarrow$ Effector (Muscles) $\longrightarrow$ Response (any other example)	1 2	3
31	(i) Hypermetropia or Far-sightedness. Reason – Image is formed behind the retina. / Near point for the person is farther away from the normal near point (25 cm)	$\frac{1}{2}$ $\frac{1}{2}$	

	(ii) - Focal length of the eye lens is too long. - The eyeball has become too small. (iii)  N = Near point of a hypermetropic eye N' = Near point of a normal eye	$\frac{1}{2}$ $\frac{1}{2}$	
32	(i) - Right - Hand Thumb Rule - If the wire carrying current is held in our right hand such that the Thumb points towards the Direction of Current, then the fingers wrap around the conductor in the direction of field lines of the magnetic field. (ii) - Fleming's Left - Hand Rule - Stretch the thumb, forefinger and middle finger of left hand mutually perpendicular to each other, such that first finger points in the direction of Magnetic Field, second finger in the direction of Current, then thumb in the direction of motion or force acting on the conductor.	$\frac{1}{2}$ 1 $\frac{1}{2}$ 1	3

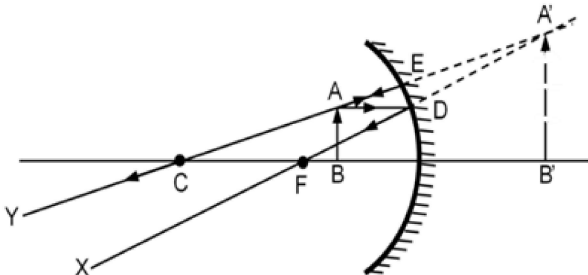
33	(A) - Number of plants/organisms of first trophic level will increase. - Number of lions/ organisms of third trophic level will decrease. - No - As the organisms of that level will find alternative foods and will not starve to death / food web is more stable where other animals as prey may be available. OR (B) - Gas 'X' is Ozone - Ozone shields the surface of the earth from ultra-violet (UV) radiations from the sun. - CFCs (Chlorofluorocarbons) - Succeeded in forging an agreement to freeze CFC production at 1986 levels / Manufacturing of CFC free refrigerators	1 1 $\frac{1}{2}$ $\frac{1}{2}$ 1 1 $\frac{1}{2}$ $\frac{1}{2}$	3
SECTION D			
34	(A) (i) A series of carbon compounds in which the same functional group substitutes for hydrogen in a carbon chain / Series of compounds having same functional group and similar chemical properties. (ii) Because melting point and boiling point increase with molecular mass. (iii) Because chemical properties of organic compounds are solely determined by their functional group which remains same in a homologous series. (iv) (i) Aldehyde: Propanal $ \begin{array}{c} \text{H} \quad \text{O} \\ \quad \\ \text{H}_3\text{C} - \text{C} - \text{C} - \text{H} \\ \\ \text{H} \end{array} / \text{CH}_3\text{CH}_2\text{CHO} $ (ii) Ketone: Propanone $ \begin{array}{c} \text{H}_3\text{C} - \text{C} - \text{CH}_3 \\ \\ \text{O} \end{array} / \text{CH}_3\text{COCH}_3 $ OR	1 1 1 $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$	

	(B) (i) Ethanol Structure: $ \begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{H} - \text{C} - \text{C} - \text{OH} \quad / \quad \text{C}_2\text{H}_5\text{OH} \quad / \quad \text{CH}_3\text{CH}_2\text{OH} \\ \quad \\ \text{H} \quad \text{H} \end{array} $ (ii) Ethene is formed $ \begin{array}{ccc} \text{C}_2\text{H}_5\text{OH} & \xrightarrow{\text{Conc. H}_2\text{SO}_4, 443\text{K (Heat)}} & \text{H}_2\text{C} = \text{CH}_2 + \text{H}_2\text{O} \\ \text{Ethanol} & & \text{Ethene} \quad \text{Water} \end{array} $ [Note: Deduct ½ mark if the conditions required are not mentioned in the equation] Concentrated Sulphuric acid acts as a dehydrating agent. (iii) Ethene 	½ 1 ½ 1 ½ ½ 1	5
35	(A) (i) - Chemical Method/Oral pills Side effects: Change the hormonal balance of the body. - Barrier method / Loop / Copper-T Side effects: Irritation in uterus. - Surgical method / Fallopian tube in female is blocked; Side effects – may cause infections. (ii) (a) Fertilized egg/zygote gets implanted in the lining of uterus and starts dividing. (b) If the egg is not fertilized, the thick and spongy lining of the uterus breaks and comes out through the vagina as blood and mucous. OR	½ ½ ½ ½ 1 1	

	(B) (i)  (a) Reproductive part – Sporangia (b) Non-reproductive part – Hypha/Hyphae. - Dry slice of bread does not provide moisture and nutrients necessary for the germination and multiplication of Rhizopus. (ii) - Budding: - Hydra uses regenerative cells for reproduction. A bud develops as an outgrowth due to repeated cell division at one specific site and develop into tiny individuals. On maturation, these buds detach from the parent and become new individuals. Alternate answer: - Regeneration: - It is carried out by specialised cells. If hydra is cut or broken into many pieces, many of these pieces grow into separate individuals. [Note: Award marks for either of the processes and its explanation]	1 $\frac{1}{2}$ $\frac{1}{2}$ 1 1 1	5
36	(A) (i) - Electric power : Rate at which electrical energy is dissipated or consumed / Rate of supplying energy to maintain the flow of current through a circuit. $P = \frac{V^2}{R}$ (ii) (a) (1 unit = 1kWh) $\text{Power, } P = \frac{\text{Electrical energy consumed}}{\text{Time}}$ $= \frac{11\text{kWh}}{5\text{h}} = 2.2\text{kW or } 2200 \text{ W}$ (b) $I = \frac{P}{V}$	1 1 $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$	

	$= \frac{2200}{220} = 10\text{A}$	1/2	
	$(c) R = \frac{V^2}{P}$ $= \frac{(220)^2}{2200} = 22 \Omega$ (Alternate formula can be used) OR	1/2	
	(B) (i) $R = \rho \frac{l}{A}$	1	
	$\rho = \frac{R \times A}{l}$ $= \text{Ohm} \times \frac{(\text{metre})^2}{\text{metre}}$ $= \text{ohm metre} / \Omega\text{m}$	1	
	(ii) Here $l = 3 \text{ m}$, $A = 4 \times 10^{-7} \text{ m}^2$, $R = 60 \Omega$ $\rho = \frac{R \times A}{l}$ $= \frac{60 \times 4 \times 10^{-7}}{3}$ $= 80 \times 10^{-7} \Omega\text{m}$	1	
	(iii) - Resistivity will not change. - because Resistivity does not depend on the dimension of the conductor / It only depends on the nature of the material.	1	
		1	
			5
	SECTION E		
37	(i) Cathode – Pure copper Anode – Impure copper	1/2	
		1/2	
	(ii) Acidified Copper Sulphate; CuSO_4	1/2 + 1/2	

	(iii) (A) - Pure copper from the anode dissolves into electrolyte and an equivalent amount of pure metal from the electrolyte is deposited on cathode / At anode : $\text{Cu} \longrightarrow \text{Cu}^{++} + 2\text{e}^{-}$ At cathode : $\text{Cu}^{++} + 2\text{e}^{-} \longrightarrow \text{Cu}$ Pure - The soluble impurities go into the solution whereas insoluble impurities settle down at the bottom of the anode. [Note: Award marks if explained with a suitable labelled diagram] OR (iii) (B) In Beaker A : - The blue colour of the solution fades (or becomes colourless) - Reason – Zn is more reactive than copper In Beaker B: - No change in colour. - Reason – Silver is less reactive than Copper	1 1 $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$	4				
38	(i) - In F₁ generation, all plants were tall / No short plants were observed - No medium height plants / No halfway characteristics were observed / Only dominant parental traits were seen and not the mixture of the two. (ii) <table border="1"> <tr> <th>Dominant trait</th> <th>Recessive trait</th> </tr> <tr> <td>Single copy of dominant trait is enough to get it expressed/always expressed</td> <td>Only expressed when present in pair.</td> </tr> </table> (Any other point) (iii) (A) - Self-pollination / Self-fertilisation / Selfing of F₁ plants - Ratio – Round Yellow : Wrinkled Green 9 : 1 - Traits are inherited independently.	Dominant trait	Recessive trait	Single copy of dominant trait is enough to get it expressed/always expressed	Only expressed when present in pair.	$\frac{1}{2}$ $\frac{1}{2}$ 1 $\frac{1}{2}$ 1	
Dominant trait	Recessive trait						
Single copy of dominant trait is enough to get it expressed/always expressed	Only expressed when present in pair.						

	OR (iii) (B) If pea plants with yellow seeds are crossed with plants of green seeds, it is found that in F₁ generation all the plants have yellow seeds. When F₁ plants are self-pollinated, it is found that in F₂ generation, plants with yellow seeds and plants with green seeds are obtained. This shows that both the traits are inherited but only one trait is visible in F₁ progeny while the other remains unexpressed. [Note: Award marks if explained by taking one characteristic / Or explained the same diagrammatically]	2	
			4
39	(i) • Mirror A. • as the object is placed beyond the centre of curvature of the mirror. (ii) Same size/ Real / Inverted (Any two) (iii) (A) Nature-Virtual and erect Size-magnified  (Deduct ½ mark if direction of rays are not marked) OR (iii) (B) Here $f = -12$ cm, $u = -18$ cm, $v = ?$ Mirror formula $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$ or $\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$ $\frac{1}{v} = \frac{1}{-12} - \frac{1}{-18}$ $v = -36$ cm In front of the mirror at a distance of 36 cm from the pole of the mirror.	½ ½ ½ + ½ ½ ½ 1 ½ ½ 1	
			4

Marking Scheme
Strictly Confidential
(For Internal and Restricted use only)
Secondary School Examination, 2024
SUBJECT: SCIENCE (086) (Q.P. CODE 31/1/2)

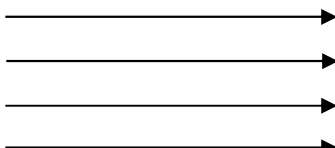
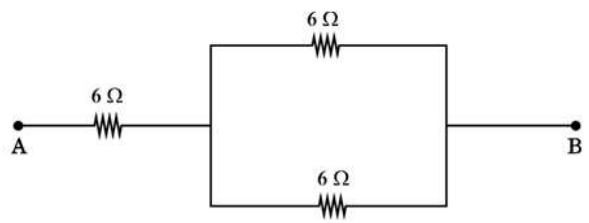
General Instructions: -

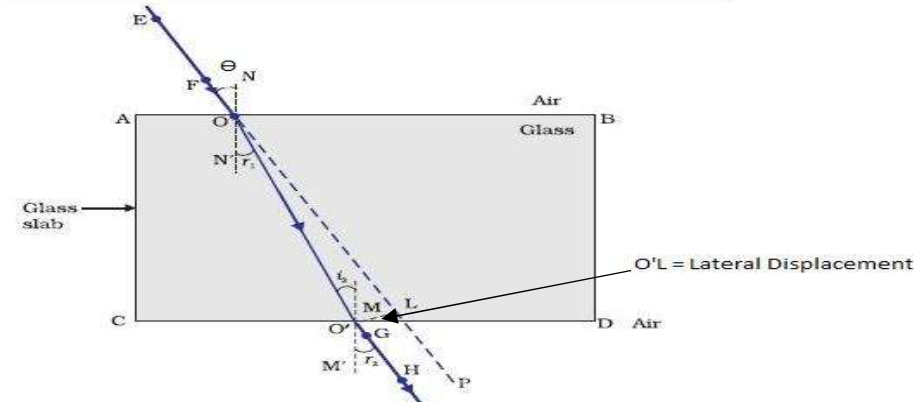
1	You are aware that evaluation is the most important process in the actual and correct assessment of the candidates. A small mistake in evaluation may lead to serious problems which may affect the future of the candidates, education system and teaching profession. To avoid mistakes, it is requested that before starting evaluation, you must read and understand the spot evaluation guidelines carefully.
2	“Evaluation policy is a confidential policy as it is related to the confidentiality of the examinations conducted, Evaluation done and several other aspects. Its’ leakage to public in any manner could lead to derailment of the examination system and affect the life and future of millions of candidates. Sharing this policy/document to anyone, publishing in any magazine and printing in News Paper/Website etc may invite action under various rules of the Board and IPC.”
3	Evaluation is to be done as per instructions provided in the Marking Scheme. It should not be done according to one’s own interpretation or any other consideration. Marking Scheme should be strictly adhered to and religiously followed. However, while evaluating, answers which are based on latest information or knowledge and/or are innovative, they may be assessed for their correctness otherwise and due marks be awarded to them. In class-X, while evaluating two competency-based questions, please try to understand given answer and even if reply is not from marking scheme but correct competency is enumerated by the candidate, due marks should be awarded.
4	The Marking scheme carries only suggested value points for the answers These are in the nature of Guidelines only and do not constitute the complete answer. The students can have their own expression and if the expression is correct, the due marks should be awarded accordingly.
5	The Head-Examiner must go through the first five answer books evaluated by each evaluator on the first day, to ensure that evaluation has been carried out as per the instructions given in the Marking Scheme. If there is any variation, the same should be zero after deliberation and discussion. The remaining answer books meant for evaluation shall be given only after ensuring that there is no significant variation in the marking of individual evaluators.
6	Evaluators will mark(√) wherever answer is correct. For wrong answer CROSS ‘X’ be marked. Evaluators will not put right (✓)while evaluating which gives an impression that answer is correct and no marks are awarded. This is most common mistake which evaluators are committing.
7	If a question has parts, please award marks on the right-hand side for each part. Marks awarded for different parts of the question should then be totaled up and written in the left-hand margin and encircled. This may be followed strictly.
8	If a question does not have any parts, marks must be awarded in the left-hand margin and encircled. This may also be followed strictly.
9	If a student has attempted an extra question, answer of the question deserving more marks should be retained and the other answer scored out with a note “Extra Question”.

10	No marks to be deducted for the cumulative effect of an error. It should be penalized only once.
11	A full scale of marks 0 – 80 (example 0 to 80/70/60/50/40/30 marks as given in Question Paper) has to be used. Please do not hesitate to award full marks if the answer deserves it.
12	Every examiner has to necessarily do evaluation work for full working hours i.e., 8 hours every day and evaluate 20 answer books per day in main subjects and 25 answer books per day in other subjects (Details are given in Spot Guidelines). This is in view of the reduced syllabus and number of questions in question paper.
13	Ensure that you do not make the following common types of errors committed by the Examiner in the past:- Leaving answer or part thereof unassessed in an answer book. Giving more marks for an answer than assigned to it. Wrong totaling of marks awarded on an answer. Wrong transfer of marks from the inside pages of the answer book to the title page. Wrong question wise totaling on the title page. Wrong totaling of marks of the two columns on the title page. Wrong grand total. Marks in words and figures not tallying/not same. Wrong transfer of marks from the answer book to online award list. Answers marked as correct, but marks not awarded. (Ensure that the right tick mark is correctly and clearly indicated. It should merely be a line. Same is with the X for incorrect answer.) Half or a part of answer marked correct and the rest as wrong, but no marks awarded.
14	While evaluating the answer books if the answer is found to be totally incorrect, it should be marked as cross (X) and awarded zero (0) Marks.
15	Any unassessed portion, non-carrying over of marks to the title page, or totaling error detected by the candidate shall damage the prestige of all the personnel engaged in the evaluation work as also of the Board. Hence, in order to uphold the prestige of all concerned, it is again reiterated that the instructions be followed meticulously and judiciously.
16	The Examiners should acquaint themselves with the guidelines given in the “Guidelines for Spot Evaluation” before starting the actual evaluation.
17	Every Examiner shall also ensure that all the answers are evaluated, marks carried over to the title page, correctly totaled and written in figures and words.
18	The candidates are entitled to obtain photocopy of the Answer Book on request on payment of the prescribed processing fee. All Examiners/Additional Head Examiners/Head Examiners are once again reminded that they must ensure that evaluation is carried out strictly as per value points for each answer as given in the Marking Scheme.

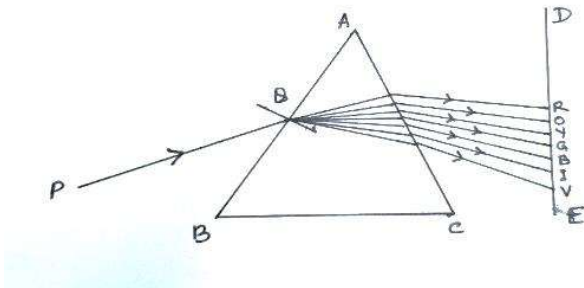
Maximum Marks: 80

X SCIENCE 31/1/2 PAGE 3

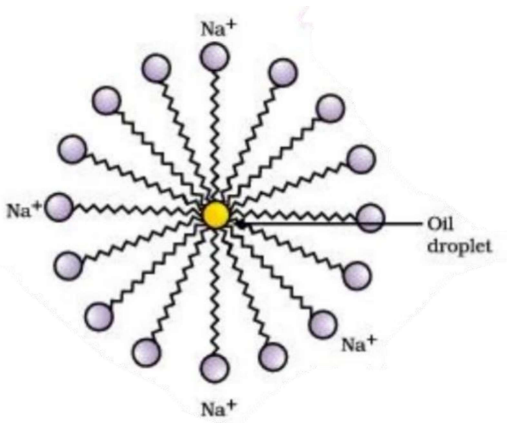
		$\frac{1}{2}$	2
22	(A) When two $6\ \Omega$ resistances are connected in parallel and the third resistance of $6\ \Omega$ is connected in series combinations to this, then equivalent resistance will be $9\ \Omega$ /  [Award marks for writing the statement or drawing the diagram] $\frac{1}{R_p} = \frac{1}{6\ \Omega} + \frac{1}{6\ \Omega}$ $\therefore R_p = 3\ \Omega$ $R_s = 6 + 3 = 9\ \Omega$ OR (B) Equivalent resistance = $R_1 + R_2 = 1\ \Omega + 2\ \Omega = 3\ \Omega$ $I = \frac{V}{R}$ $= \frac{6\ \text{V}}{1\ \Omega + 2\ \Omega} = \frac{6\ \text{V}}{3\ \Omega} = 2\ \text{A}$ Electric power, $P = I^2 R$ $= (2\ \text{A})^2 \times 2\ \Omega = 4 \times 2\ \text{W} = 8\ \text{W}$	1 1 $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$	2

23	 Lateral displacement labelling	1½	
		½	2
24	(A) Medulla – Hindbrain Function– Control blood pressure/salivation/vomiting or any other Cerebrum – Forebrain Function–Thinking/intelligence/memory (any other) OR (B)Auxins When tendril of pea plant comes in contact with any support, the part of the tendril in contact with the object does not grow as rapidly as part of the tendril away from it. This causes the tendril to circle around the object and cling to it.	½ ½ ½ ½	
		1 1	2
25	Kidney → Ureter → Urinary bladder → Urethra Reabsorption of nutrients/amino acids, glucose and water will not take place	1 1	2
26	(i) $3\text{BaCl}_2(\text{aq}) + \text{Al}_2(\text{SO}_4)_3(\text{aq}) \longrightarrow 3\text{BaSO}_4(\text{s}) + 2\text{AlCl}_3(\text{aq})$ Equation Balancing (ii) $2\text{Al}(\text{s}) + 3\text{H}_2\text{O}(\text{g}) \longrightarrow \text{Al}_2\text{O}_3(\text{s}) + 3\text{H}_2(\text{g})$ Equation Balancing	½ ½ ½ ½	2

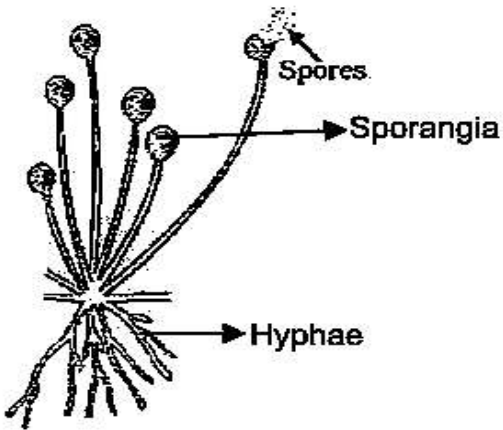
	SECTION C		
27	(i) The taste of tomato juice will be slightly sour; The pH 4.6 indicates that tomato juice is an acid and acids are sour in taste. (ii) Acids that give more H⁺ ions / H₃O⁺ are Strong Acids Bases that give less OH⁻ ions are Weak Bases. (iii) Living animals can survive within a pH range of 7.0 to 7.8. So, if the pH of river water becomes low due to acid rain (pH < 5.6), then survival of aquatic animals becomes difficult.	½ ½ ½ ½ 1	3
28	(i) Change in colour: The solution will become green in colour. $\begin{array}{ccccccc} \text{Fe(s)} & + & \text{CuSO}_4(\text{aq}) & \longrightarrow & \text{FeSO}_4 & + & \text{Cu(s)} \\ & & \text{Blue} & & \text{Green} & & \end{array}$ (or any other reaction which shows change in colour) (ii) Change in temperature: The temperature will increase. $\text{NaOH(aq)} + \text{HCl(aq)} \longrightarrow \text{NaCl(aq)} + \text{H}_2\text{O(l)} + \text{Heat}$ (or any other reaction which shows change in temperature) (iii) Formation of precipitate: Yellow precipitate of PbI₂ is formed. $\text{Pb(NO}_3)_2(\text{aq}) + 2\text{KI(aq)} \longrightarrow \text{PbI}_2(\text{s}) + 2\text{KNO}_3(\text{aq})$ Yellow (or any other reaction which shows formation of precipitate)	½ ½ ½ ½ ½ ½	3
29	Reflex action is a sudden/spontaneous/immediate action in response to the environment/stimulus e.g. sneezing. Stimulus → Receptors (Nose) → Sensory neuron Response ← Effector (Muscles) ← Motor neuron ← Spinal cord ← (Relay neuron) (any other example)	1 2	3
30	(i) Leaves with green (Chlorophyll) and non-green patches (white or yellow patches) e.g. croton/money plant/ any other (ii) Leaf becomes colorless;	½ ½ ½	

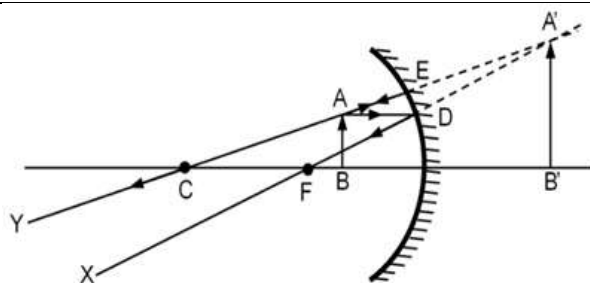
	The solution becomes green (iii) Starch. Chlorophyll helps the plant to absorb energy of the sunlight for the process of photosynthesis	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$	3
31	(A) - Number of plants/organisms of first trophic level will increase. - Number of lions/ third trophic level will decrease. - No - As the organisms of that level will find alternative foods and will not starve to death / food web is more stable where other animals as prey may be available. OR (B) - Gas 'X' is Ozone - Ozone shields the surface of the earth from ultra-violet (UV) radiations from the sun. - CFCs (Chlorofluorocarbons) - Succeeded in forging an agreement to freeze CFC production at 1986 levels / Manufacturing of CFC free refrigerators	1 1 $\frac{1}{2}$ $\frac{1}{2}$ 1 1 $\frac{1}{2}$ $\frac{1}{2}$	3
32	 - Phenomenon: Dispersion of light - Cause : Different colours of white light bend through different angles with respect to incident ray./ Different colours of white light have different wavelength therefore bend by different angles. - Refractive index of glass is highest for violet colour. - Justification :For same $\angle i$ the $\angle r$ is minimum for the violet light. $\left(\frac{\sin i}{\sin r}\right)$ or refractive index is highest	1 $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$	3
33	(i) • Electric fuse and Earth wire (ii) $I = P/V$	$\frac{1}{2} + \frac{1}{2}$ $\frac{1}{2}$	

	$= \frac{2000 \text{ W}}{220 \text{ V}} = 9.09 \text{ A}$ Since, current drawn by the oven is greater than the rated value of current, (9.09A > 5A), the fuse wire melts/ the electric oven stops working.	$\frac{1}{2}$ 1	3
	SECTION-D		
34	(A) (i) Functional Group: A hetero atom or group of atoms attached to the carbon chain, which gives specific properties to the carbon compounds. (I) Ketone (II) Carboxylic acid (ii) Ethanoic acid is formed $\text{C}_2\text{H}_5\text{OH} \xrightarrow{\text{Acidified K}_2\text{Cr}_2\text{O}_7 + \text{Heat}} \text{CH}_3\text{COOH}$ • oxygen is added to ethanol and converts /oxidises ethanol to ethanoic acid. (iii) $\text{CH}_3\text{COOH} + \text{NaOH} \longrightarrow \text{CH}_3\text{COONa} + \text{H}_2\text{O}$ OR (B) (i) Soaps are prepared by heating an ester (animal fat / vegetable oil) with a base such as sodium hydroxide. $\text{CH}_3\text{COOC}_2\text{H}_5 \xrightarrow{\text{NaOH}} \text{CH}_3\text{COONa} + \text{C}_2\text{H}_5\text{OH}$ Sodium ethanoate (ii) Ionic (hydrophilic) end of the soap interacts with water while the carbon chain(hydrophobic) interacts with oil. Thus micelles are formed. Emulsion is formed in the water. Soap micelles pull out the dirt and oil in water.	1 $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ 1 $\frac{1}{2}$ 1 1 2	

		1	5
35	(A) (i) - Electric power : Rate at which electrical energy is dissipated or consumed / Rate of supplying energy to maintain the flow of current through a circuit. $P = \frac{V^2}{R}$ (ii) (a) (1 unit = 1kWh) $\text{Power, } P = \frac{\text{Electrical energy consumed}}{\text{Time}}$ $= \frac{11\text{kWh}}{5\text{h}} = 2.2\text{kW or } 2200 \text{ W}$ (b) $I = \frac{P}{V}$ $= \frac{2200}{220} = 10\text{A}$ (c) $R = \frac{V^2}{P}$ $= \frac{(220)^2}{2200} = 22 \Omega$ (Alternate formula can be used) OR	1 1 $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$	

	(B) (i) $R = \rho \frac{l}{A}$ $\rho = \frac{R \times A}{l}$ $= \text{Ohm} \times \frac{(\text{metre})^2}{\text{metre}}$ $= \text{ohm meter} / \Omega\text{m}$ (ii) Here $l = 3 \text{ m}$, $A = 4 \times 10^{-7} \text{ m}^2$, $R = 60 \Omega$ $\rho = \frac{R \times A}{l}$ $= \frac{60 \times 4 \times 10^{-7}}{3}$ $= 80 \times 10^{-7} \Omega\text{m}$ (iii) - Resistivity will not change. - because Resistivity does not depend on the dimension of the conductor / It only depends on the nature of the material.	1 1 1 1 1	5
36	(A) (i) - Chemical Method/Oral pills Side effects: Change the hormonal balance of the body. - Barrier method / Loop / Copper-T Side effects: Irritation in uterus. - Surgical method / Fallopian tube in female is blocked; Side effects – may cause infections. (ii) (a) Fertilized egg/zygote gets implanted in the lining of uterus and starts dividing. (b) If the egg is not fertilized, the thick and spongy lining of the uterus breaks and comes out through the vagina as blood and mucous. OR	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ 1 1	

	(B) (i)  (a) Reproductive part – Sporangia (b) Non-reproductive part – Hypha/Hyphae. - Dry slice of bread does not provide moisture and nutrients necessary for the germination and multiplication of Rhizopus. (ii) - Budding: - Hydra uses regenerative cells for reproduction. A bud develops as an outgrowth due to repeated cell division at one specific site and develop into tiny individuals. On maturation, these buds detach from the parent and become new individuals. Alternate answer: - Regeneration: - It is carried out by specialised cells. If hydra is cut or broken into many pieces, many of these pieces grow into separate individuals. [Note: Award marks for either of the processes and its explanation]	1 $\frac{1}{2}$ $\frac{1}{2}$ 1 1 1	5
	SECTION E		
37	(i) - Mirror A. - as the object is placed beyond the centre of curvature of the mirror. (ii) Same size/ Real / Inverted (Any two) (iii) (A) Nature-Virtual and erect Size-magnified	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2} + \frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$	



(Deduct ½ mark if direction of rays are not marked)

OR

(iii) (B) Here $f = -12$ cm, $u = -18$ cm, $v = ?$

Mirror formula $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$ or $\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$

$$\frac{1}{v} = \frac{1}{-12} - \frac{1}{-18}$$

$$v = -36\text{cm}$$

In front of the mirror at a distance of 36 cm from the pole of the mirror.

1

½

½

1

4

38

(i)

- In F_1 generation, all plants were tall / No short plants were observed
- No medium height plants / No halfway characteristics were observed / Only dominant parental traits were seen and not the mixture of the two.

½

½

(ii)

Dominant trait	Recessive trait
Single copy of dominant trait is enough to get it expressed/always expressed	Only expressed when present in pair.

1

(Any other point)

(iii) (A)

- Self-pollination / Self-fertilisation/ Selfing of F_1 plants
- Ratio – Round Yellow : Wrinkled Green
9 : 1
- Traits are inherited independently.

½

½

1

	OR (iii) (B) If pea plants with yellow seeds are crossed with plants of green seeds, it is found that in F₁ generation all the plants have yellow seeds. When F₁ plants are self-pollinated, it is found that in F₂ generation, plants with yellow seeds and plants with green seeds are obtained. This shows that both the traits are inherited but only one trait is visible in F₁ progeny while the other remains unexpressed. [Note: Award marks if explained by taking one characteristic / Or explained the same diagrammatically]	2	4
39	(i) Cathode – Pure copper Anode – Impure copper (ii) Acidified Copper Sulphate; CuSO₄ (iii) (A) - Pure copper from the anode dissolves into electrolyte and an equivalent amount of pure metal from the electrolyte is deposited on cathode / At anode : $\text{Cu} \longrightarrow \text{Cu}^{++} + 2\text{e}^{-}$ At cathode : $\text{Cu}^{++} + 2\text{e}^{-} \longrightarrow \text{Cu}$ Pure - The soluble impurities go into the solution whereas insoluble impurities settle down at the bottom of the anode. [Note: Award marks if explained with a suitable labelled diagram] OR (iii) (B) In Beaker A : • The blue colour of the solution fades (or becomes colourless) • Reason – Zn is more reactive than copper In Beaker B: • No change in colour. • Reason – Silver is less reactive than Copper	½ ½ ½ + ½ 1 1 ½ ½ ½ ½	4

Marking Scheme
Strictly Confidential
Secondary School Examination, 2024
SUBJECT NAME SCIENCE (086) (Q.P. CODE 31/1/3)

General Instructions: -

1	You are aware that evaluation is the most important process in the actual and correct assessment of the candidates. A small mistake in evaluation may lead to serious problems which may affect the future of the candidates, education system and teaching profession. To avoid mistakes, it is requested that before starting evaluation, you must read and understand the spot evaluation guidelines carefully.
2	“Evaluation policy is a confidential policy as it is related to the confidentiality of the examinations conducted, Evaluation done and several other aspects. Its’ leakage to public in any manner could lead to derailment of the examination system and affect the life and future of millions of candidates. Sharing this policy/document to anyone, publishing in any magazine and printing in News Paper/Website etc may invite action under various rules of the Board and IPC.”
3	Evaluation is to be done as per instructions provided in the Marking Scheme. It should not be done according to one’s own interpretation or any other consideration. Marking Scheme should be strictly adhered to and religiously followed. However, while evaluating, answers which are based on latest information or knowledge and/or are innovative, they may be assessed for their correctness otherwise and due marks be awarded to them. In class-X, while evaluating two competency-based questions, please try to understand given answer and even if reply is not from marking scheme but correct competency is enumerated by the candidate, due marks should be awarded.
4	The Marking scheme carries only suggested value points for the answers. These are in the nature of Guidelines only and do not constitute the complete answer. The students can have their own expression and if the expression is correct, the due marks should be awarded accordingly.
5	The Head-Examiner must go through the first five answer books evaluated by each evaluator on the first day, to ensure that evaluation has been carried out as per the instructions given in the Marking Scheme. If there is any variation, the same should be zero after deliberation and discussion. The remaining answer books meant for evaluation shall be given only after ensuring that there is no significant variation in the marking of individual evaluators.
6	Evaluators will mark(√) wherever answer is correct. For wrong answer CROSS ‘X’ be marked. Evaluators will not put right (√)while evaluating which gives an impression that answer is correct and no marks are awarded. This is most common mistake which evaluators are committing.
7	If a question has parts, please award marks on the right-hand side for each part. Marks awarded for different parts of the question should then be totaled up and written in the left-hand margin and encircled. This may be followed strictly.
8	If a question does not have any parts, marks must be awarded in the left-hand margin and encircled. This may also be followed strictly.
9	If a student has attempted an extra question, answer of the question deserving more marks should be retained and the other answer scored out with a note “Extra Question”.

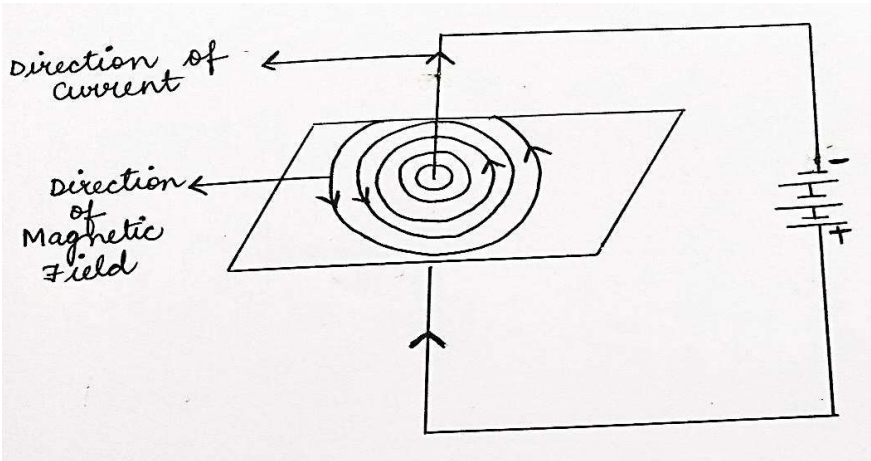
10	No marks to be deducted for the cumulative effect of an error. It should be penalized only once.
11	A full scale of marks 0 – 80 (example 0 to 80/70/60/50/40/30 marks as given in Question Paper) has to be used. Please do not hesitate to award full marks if the answer deserves it.
12	Every examiner has to necessarily do evaluation work for full working hours i.e., 8 hours every day and evaluate 20 answer books per day in main subjects and 25 answer books per day in other subjects (Details are given in Spot Guidelines). This is in view of the reduced syllabus and number of questions in question paper.
13	Ensure that you do not make the following common types of errors committed by the Examiner in the past:- Leaving answer or part thereof unassessed in an answer book. Giving more marks for an answer than assigned to it. Wrong totaling of marks awarded on an answer. Wrong transfer of marks from the inside pages of the answer book to the title page. Wrong question wise totaling on the title page. Wrong totaling of marks of the two columns on the title page. Wrong grand total. Marks in words and figures not tallying/not same. Wrong transfer of marks from the answer book to online award list. Answers marked as correct, but marks not awarded. (Ensure that the right tick mark is correctly and clearly indicated. It should merely be a line. Same is with the X for incorrect answer.) Half or a part of answer marked correct and the rest as wrong, but no marks awarded.
14	While evaluating the answer books if the answer is found to be totally incorrect, it should be marked as cross (X) and awarded zero (0) Marks.
15	Any unassessed portion, non-carrying over of marks to the title page, or totaling error detected by the candidate shall damage the prestige of all the personnel engaged in the evaluation work as also of the Board. Hence, in order to uphold the prestige of all concerned, it is again reiterated that the instructions be followed meticulously and judiciously.
16	The Examiners should acquaint themselves with the guidelines given in the “Guidelines for Spot Evaluation” before starting the actual evaluation.
17	Every Examiner shall also ensure that all the answers are evaluated, marks carried over to the title page, correctly totaled and written in figures and words.
18	The candidates are entitled to obtain photocopy of the Answer Book on request on payment of the prescribed processing fee. All Examiners/Additional Head Examiners/Head Examiners are once again reminded that they must ensure that evaluation is carried out strictly as per value points for each answer as given in the Marking Scheme.

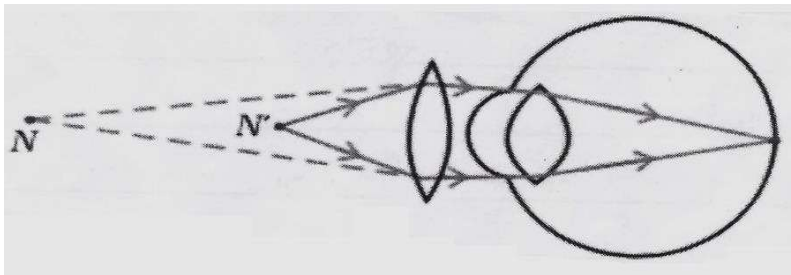
MARKING SCHEME
Secondary School Examination, 2024
SCIENCE (Subject Code–086)
[Paper Code: 31/1/3]

Maximum Marks: 80

Q. No.	EXPECTED ANSWER / VALUE POINTS	Marks	Total Marks
SECTION A			
1	(c) / $2 \text{ AgBr} \longrightarrow 2 \text{ Ag} + \text{Br}_2$	1	1
2	(b) / $2 \text{ NaOH} + \text{Zn} \longrightarrow \text{Na}_2\text{ZnO}_2 + \text{H}_2$	1	1
3	(d) / MnO_2 is reduced and HCl is oxidised	1	1
4	(d) / Na_2CO_3	1	1
5	(c) / Neuromuscular junction	1	1
6	(c) / Mercury and Bromine	1	1
7	(c) / At twice the focal length of the lens	1	1
8	(c) / (ii) and (iv)	1	1
9	(c) / amphoteric	1	1
10	(d) / (i) and (iv)	1	1
11	(b) / (ii) and (iv)	1	1
12	(c) / Vas deferens	1	1
13	(d) / Plasmodium	1	1
14	(d) / The upper portion is of concave lens for the distant vision and lower part is of convex lens for the near vision.	1	1
15	(a) / 	1	1
16	(c) / Tiger, grass, snake, frog	1	1
17	(c) / Assertion (A) is true, but Reason (R) is false.	1	1
18	(a) / Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).	1	1
19	(b) / Both Assertion (A) and Reason (R) are true, but Reason (R) is <i>not</i> the correct explanation of Assertion (A).	1	1
20	(c) / Assertion (A) is true, but Reason (R) is false.	1	1
SECTION-B			
21	(A) (i) The communication between the central nervous system and the other parts of the body is facilitated by the peripheral nervous systems. (ii) protected in a bony box/skull//cranium/fluid filled balloon like structure which provides shock absorption. OR	1 1	

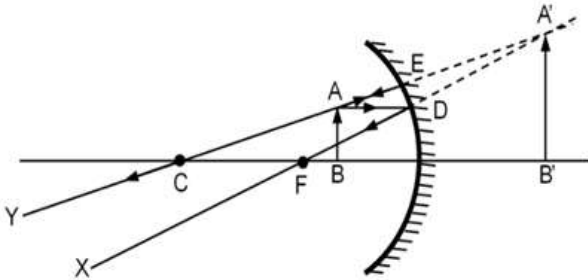
	(B) Chemotropism ; eg. growth of pollen tubes towards the ovules. Hydrotropism ; eg. growth of roots towards water.	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$	2					
22	(i) Herbivores eating grass need a longer small intestine to allow the cellulose to be digested. Meat is easier to digest. Hence carnivores have shorter small intestine. (ii) <table><tr><th>Pepsin</th><th>Trypsin</th></tr><tr><td>i. Secreted by the gastric glands present in the walls of stomach</td><td>Secreted in pancreas</td></tr><tr><td>ii. Acts in acidic medium</td><td>Acts in alkaline medium</td></tr></table> (Any one)	Pepsin	Trypsin	i. Secreted by the gastric glands present in the walls of stomach	Secreted in pancreas	ii. Acts in acidic medium	Acts in alkaline medium	1 <
Pepsin	Trypsin							
i. Secreted by the gastric glands present in the walls of stomach	Secreted in pancreas							
ii. Acts in acidic medium	Acts in alkaline medium							

25	 Direction of current Direction of Magnetic Field Lines	1	
		$\frac{1}{2}$ $\frac{1}{2}$	2
26	$u = -10\text{ cm}; f = +15\text{ cm}$ $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$ $\frac{1}{15} = \frac{1}{v} + \frac{1}{-10\text{ cm}}$ $\frac{1}{v} = \frac{1}{15\text{ cm}} + \frac{1}{10\text{ cm}}$ $v = +6\text{ cm}$ Image is formed behind the mirror.	$\frac{1}{2}$ $\frac{1}{2}$	
		1	2
SECTION-C			
27	(A) • Number of plants/organisms of first trophic level will increase. • Number of lions/ third trophic level will decrease. • No • As the organisms of that level will find alternative foods and will not starve to death / food web is more stable where other animals as prey may be available. OR (B) • Gas 'X' is Ozone • Ozone shields the surface of the earth from ultra-violet (UV) radiations from the sun. • CFCs (Chlorofluorocarbons)	1 1 $\frac{1}{2}$ $\frac{1}{2}$	
		1 1 $\frac{1}{2}$	

	Succeeded in forging an agreement to freeze CFC production at 1986 levels / Manufacturing of CFC free refrigerators	1/2	3
28	(i) - Right - Hand Thumb Rule - If the wire carrying current is held in our right hand such that the Thumb points towards the Direction of Current, then the fingers wrap around the conductor in the direction of field lines of the magnetic field. (ii) - Fleming's Left - Hand Rule - Stretch the thumb, forefinger and middle finger of left hand mutually perpendicular to each other, such that first finger points in the direction of Magnetic Field, second finger in the direction of Current, then thumb in the direction of motion or force acting on the conductor.	1/2 1 1/2 1	3
29	(i) Hypermetropia or Far-sightedness. Reason – Image is formed behind the retina. / Near point for the person is farther away from the normal near point (25 cm) (ii) - Focal length of the eye lens is too long. - The eyeball has become too small. (iii)  N = Near point of a hypermetropic eye N' = Near point of a normal eye	1/2 1/2 1/2 1/2 1	3
30	Reflex action is a sudden/spontaneous/immediate action in response to the environment/stimulus e.g. sneezing. <pre> Stimulus → Receptors (Nose) → Sensory neuron → Spinal cord (Relay neuron) → Motor neuron → Effector (Muscles) → Response </pre> (any other example)	1 2	3
31	(i)Amphibians - frogs / Reptiles - lizards	1/2	

	- The body temperature depends on the temperature in the environment. Therefore they can tolerate some mixing of the oxygenated and de-oxygenated blood streams. (ii) Two functions: - Lymph carries digested and absorbed fat from intestine - Drains excess fluid from extra cellular space back into the blood.	$\frac{1}{2}$ 1 1	3
32	(i) Plaster of Paris ; Calcium Sulphate hemihydrate (ii) $CaSO_4 \cdot 2 H_2O \xrightarrow{373 K} CaSO_4 \cdot \frac{1}{2} H_2O + 1 \frac{1}{2} H_2O$ (iii) Two uses : - Used for making toys - Materials for decoration - Making surfaces smooth - Supporting fractured bones (Any two) (Any other alternate answer)	$\frac{1}{2} + \frac{1}{2}$ 1 $\frac{1}{2} + \frac{1}{2}$	3
33	(i) A reaction in which a single substance on absorption of energy. decomposes to give two or more substances. $2 Pb(NO_3)_2 \xrightarrow{Heat} 2 PbO + 4 NO_2 + O_2$ (ii) Cathode : Anode (Mass ratio) 1 : 8	1 1 1	3
SECTION-D			
34	(A)(i)(a) same current and same potential difference. (b) same current and same potential difference (c) same current but different potential difference (d) different current but same potential difference. (ii) (a) Minimum resistance – When resistors are in parallel $\frac{1}{R} = \frac{1}{24} + \frac{1}{24}$	$\frac{1}{2} \times 4$ $\frac{1}{2}$	

	(i) A series of carbon compounds in which the same functional group substitutes for hydrogen in a carbon chain / Series of compounds having same functional group and similar chemical properties. (ii) Because melting point and boiling point increase with molecular mass. (iii) Because chemical properties of organic compounds are solely determined by their functional group which remains same in a homologous series. (iv) (i) Aldehyde: Propanal $ \begin{array}{c} \text{H} \quad \text{O} \\ \quad \\ \text{H}_3\text{C} - \text{C} - \text{C} - \text{H} \\ \\ \text{H} \end{array} \quad / \text{CH}_3\text{CH}_2\text{CHO} $ (ii) Ketone: Propanone $ \begin{array}{c} \text{H}_3\text{C} - \text{C} - \text{CH}_3 \\ \\ \text{O} \end{array} \quad / \text{CH}_3\text{COCH}_3 $ OR (B) (i) Ethanol Structure: $ \begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{H} - \text{C} - \text{C} - \text{OH} \\ \quad \\ \text{H} \quad \text{H} \end{array} \quad / \text{C}_2\text{H}_5\text{OH} / \text{CH}_3\text{CH}_2\text{OH} $ (ii) Ethene is formed $ \begin{array}{ccc} \text{C}_2\text{H}_5\text{OH} & \xrightarrow{\text{Conc.H}_2\text{SO}_4, 443\text{K (Heat)}} & \text{H}_2\text{C} = \text{CH}_2 + \text{H}_2\text{O} \\ \text{Ethanol} & & \text{Ethene} \quad \text{Water} \end{array} $ [Note: Deduct ½ marks if the conditions required are not mentioned in the equation] Concentrated Sulphuric acid acts as a dehydrating agent.	1 1 1 ½ ½ ½ ½ ½ 1 ½ 1 ½	
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	plants are self-pollinated, it is found that in F₂ generation, plants with yellow seeds and plants with green seeds are obtained. This shows that both the traits are inherited but only one trait is visible in F₁ progeny while the other remains unexpressed. [Note: Award marks if explained by taking one characteristic / Or explained the same diagrammatically]	2	4
38	(i) • Mirror A. • as the object is placed beyond the centre of curvature of the mirror. (ii) Same size/ Real / Inverted (Any two) (iii) (A) Nature-Virtual and erect Size-magnified  (Deduct ½ mark if direction of rays are not marked) OR (iii) (B) Here $f = -12$ cm, $u = -18$ cm, $v = ?$ Mirror formula $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$ or $\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$ $\frac{1}{v} = \frac{1}{-12} - \frac{1}{-18}$ $v = -36$ cm In front of the mirror at a distance of 36 cm from the pole of the mirror.	½ ½ ½ + ½ ½ ½ 1 ½ ½ 1	4
39	(i) Cathode – Pure copper Anode – Impure copper	½ ½	

	(ii) Acidified Copper Sulphate; CuSO_4 (iii) (A) - Pure copper from the anode dissolves into electrolyte and an equivalent amount of pure metal from the electrolyte is deposited on cathode / At anode : $\text{Cu} \longrightarrow \text{Cu}^{++} + 2\text{e}^-$ At cathode : $\text{Cu}^{++} + 2\text{e}^- \longrightarrow \text{Cu}$ Pure - The soluble impurities go into the solution whereas insoluble impurities settle down at the bottom of the anode. [Note: Award marks if explained with a suitable labelled diagram] OR (iii) (B) In Beaker A : • The blue colour of the solution fades (or becomes colourless) • Reason – Zn is more reactive than copper In Beaker B: • No change in colour. • Reason – Silver is less reactive than Copper	$\frac{1}{2} + \frac{1}{2}$ 1 1 $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$	4
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